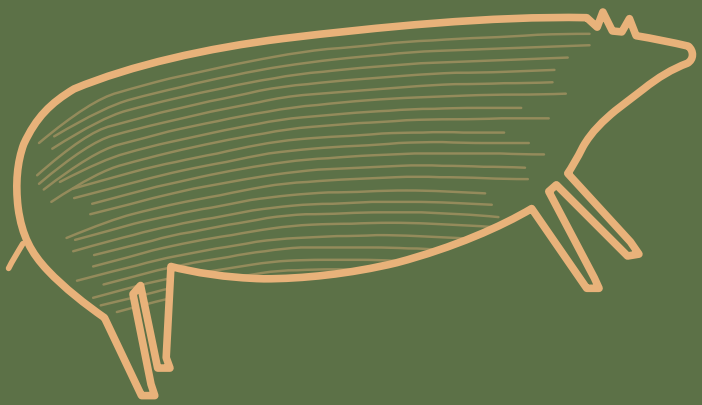
DRAFT VERSION 3 FOR REVIEW · 5 OCTOBER 2026

AN EVIDENCE GUIDE TO HUMAN ORIGINS

How do we know humans evolved?

Fossils. Ancient footprints. DNA. A chromosome with a telltale join. Key claims with a number, a stated uncertainty and a source you can check.



Cover: one of the oldest known animal paintings by humans, a warty pig in Leang Tedongnge cave, Sulawesi, Indonesia, at least 45,500 years old.

George Philippou
Human-directed, AI-assisted, source-checked
Draft version 3 for review · 5 October 2026

Evidence checked against the Natural History Museum, London, and the Smithsonian Institution

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How this booklet makes its case

Parts 1–6 set out the evidence, and each page answers one question. Part 7 asks what the same method means for claims resting on revelation. Each part has its own colour, shown on the edge of every page.

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1

PART 1

How science knows

How can we know about a past no one saw?

This booklet asks one question: how do we know humans evolved? First, the tool used to answer it: one scientific method, trusted everywhere, shown at work on three ages, the universe, the Earth and life, each with its number, its error bar and a way it could be wrong. The result: there has been time for evolution. Part 2 shows how it works.

IN THIS PART

4 Why does one method apply everywhere? · Science is one method, not a menu
5 What is a scientific theory? · A theory is a tested explanation, not a guess
6 How old is the universe? · About 13.8 billion years
7 How old is the Earth? · About 4.54 billion years
8 When did life begin? · At least 3.4 billion years ago

WHY DOES ONE METHOD APPLY EVERYWHERE?

Science is one method, not a menu

Evolution is not a separate kind of science with its own rules.

The same scientific principles we rely on every day also explain the history of life, as the four pictures below show. **Dating** reads radioactive decay: unstable atoms break down at a fixed rate, the same nuclear physics that runs power stations and hospital scanners. **Genetics** reads the paired letters of DNA, just as it does to match organ donors and diagnose inherited disease. **Geology** reads rock layers and the faults that break them, which is how it finds oil and maps earthquake risk. **Physics** keeps satellite navigation working: the clocks on the satellites run fast by about 38 millionths of a second a day, as relativity predicts, and must be corrected. [1] The same physics gives the universe an age.

These sciences use different tools, from telescopes to gene sequencers, but share one logic: observe, measure, predict, test, and let the evidence decide. The logic that keeps hospitals and power stations running is the one this booklet rests on. [2, 3]

Science does not answer every question, or say what a life means. But the physical history of our species is a question of evidence, and asking how a claim could be checked, then holding a belief in proportion to the evidence, is ordinary due diligence. The rest of this part shows the method on three older questions, each set out the same way: **the method**, **the number** and **what could be wrong**.

How this booklet was checked. The evidence on pages 6–48 has been checked against the Natural History Museum (NHM, London), the Smithsonian Human Origins Program (Washington, DC) and the US National Academy of Sciences. It also rests on the peer-reviewed research listed on pages 62–66. Page 60 shows which museum page backs each claim. Sources were checked in September 2026; the newest findings are the likeliest to be refined. None of these institutions produced or endorsed this booklet.



Dating
an atom decaying
power, hospital scans



Genetics
DNA's paired letters
donors, disease



Geology
layers cut by a fault
oil, earthquake risk



Physics
a satellite's clock
phone navigation

the same method: observe · measure · predict · test

WHAT IS A SCIENTIFIC THEORY?

A theory is a tested explanation, not a guess

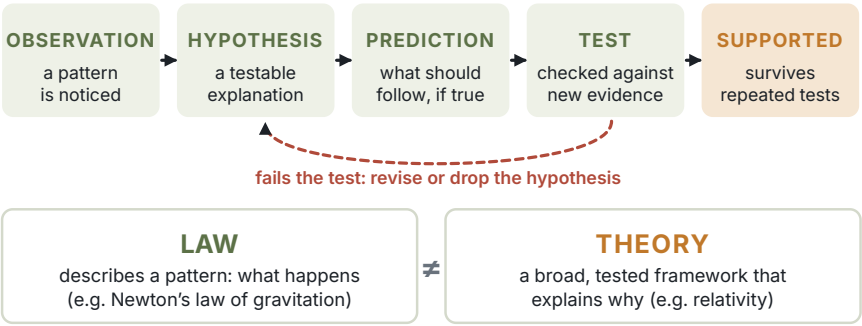
A scientific theory is not a guess. It is an explanation that has survived repeated attempts to prove it wrong.

Science works by testing ideas against the world, in the cycle shown below. A *hypothesis* is a testable explanation, precise enough to be wrong. From it comes a prediction: if the hypothesis is true, a particular new observation should follow. If the prediction fails, the hypothesis is revised or dropped; if it survives repeated, independent tests, it is supported. A *theory* is larger: a broad, well-tested framework that ties many facts and hypotheses together. It is not a guess waiting for promotion, but the strongest kind of explanation science produces. [4]

A "law" is not a more advanced theory. A law describes a pattern, often as an equation: Newton's law says how gravity depends on mass and distance. A theory explains *why*: general relativity explains gravity as mass curving space and time. A theory never "graduates" into a law. [4]

Fact and theory are not rivals. A *fact* is something observed, or confirmed so thoroughly that there is no longer good reason to doubt it. That things fall is a fact; the theory of gravity explains why. Evolution is both: that populations change, and that humans have evolutionary ancestors, are facts; the theory of evolution explains how. [2, 3]

Germ theory, atomic theory and plate tectonics are theories in exactly this sense, and they are among the best-supported explanations science has. [2, 4]



CHECK IT UC Berkeley, *Understanding Science* **[4]** · US National Academies **[2, 3]**

HOW OLD IS THE UNIVERSE?

About 13.8 billion years

The universe has been expanding from a hot, dense early state for about 13.8 billion years, a figure measured to within about 20 million years.

The method. The universe is expanding: the farther away a galaxy is, the faster it moves away. Run that backwards, and the universe was once extremely hot and dense. The most precise measurement of its age comes from the *cosmic microwave background*, a faint glow released when the young universe first became transparent, about 380,000 years after the start. The tiny temperature ripples in that glow, pictured below, depend on how much matter the universe holds, how fast it has expanded and for how long.

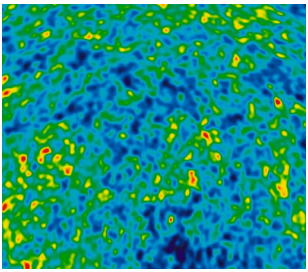
The number. The Planck satellite mapped those ripples across the whole sky. Fitting the standard model of the universe to that map gives **13.8 billion years, ± about 20 million**: roughly one part in seven hundred. [5]

What could be wrong. The age depends on the model, so the model is checked in other ways: the oldest stars, for example, must be younger than the universe, and they are. Two ways of measuring today's expansion rate currently disagree. The quoted error bar holds within the standard model; a different model, or settling that disagreement, could shift the age.

13.8 bn
years: the age of the universe

± 20 million
years: the stated uncertainty

380,000
years after the start: the glow is released



A patch of the real microwave sky. Hot and cold spots differ by about 0.0001 °C. NASA / WMAP, public domain.



CHECK IT Planck Collaboration 2020 [5]

HOW OLD IS THE EARTH?

About 4.54 billion years

The Earth formed about 4.54 billion years ago, measured with uranium clocks that check each other.

The method. Some atoms decay into others at fixed rates, which have been measured for over a century. Uranium decays into lead by two routes, at different speeds. A zircon crystal takes in uranium when it forms, but almost no lead, so the lead inside it now was made by decay. Each crystal holds two clocks, and they must agree (diagram below).

The number. No rock survives from the Earth's formation, but meteorites do: they are leftovers from the disc of dust that formed the planets. Measurements on meteorites since 1956 give **4.54 billion years, ± about 50 million**. [6] The Earth's oldest known fragments, zircons from Jack Hills in Western Australia, date to **4.4 billion years** by the same method. [7]

What could be wrong. If decay rates had changed, these clocks would mislead. But the rates have held steady in every test, and meteorites, Moon rocks and the Earth's oldest minerals give consistent ages.

THE SAME SCIENCE, IN DAILY USE

Hospitals rely on the same fixed decay rates. Technetium-99m, used in tens of millions of scans a year, loses half its radioactivity every six hours, and doses are timed by it. [8]

4.54 bn
years: the age of the Earth

± 50 million
years: the stated uncertainty

4.4 bn
years: the oldest known zircon



A zircon crystal (illustrative). Rob Lavinsky, iRocks.com, CC BY-SA 3.0.

TWO CLOCKS IN ONE CRYSTAL — *shown at 4.54 billion years*

uranium-238 → lead-206 — half-life 4.5 billion years

about half the uranium left | lead made by decay

uranium-235 → lead-207 — half-life 0.7 billion years

about 1% of the uranium left, the rest now lead

Each ratio gives an age, and the two must agree. Measured this way:

4.54 bn yr	meteorites (age of the Earth)	4.4 bn yr	Jack Hills zircon (oldest crystal)

CHECK IT NHM: *Origin of life on Earth* [9] · Dalrymple 2001 [6] · National Academies 2016 [8]

WHEN DID LIFE BEGIN?

At least 3.4 billion years ago

1

Microbes were building reefs at least 3.43 billion years ago.

The method. Mats of microbes in shallow seas trap sediment and build layered mounds called stromatolites; they still form today. Ancient rocks in the Pilbara region of Western Australia contain them (the photograph below is a slightly older example from the same region). A 2006 study mapped them over ten kilometres: their shapes, variety and setting match a living reef, not any known chemical process. [10]

The number. The rocks are about **3.43 billion years old**. Microbial life was present, and widespread, by that time. [10]

What could be wrong. Claims of older life are less secure. In 2016 a team reported similar structures in 3.7-billion-year-old rocks at Isua, Greenland. [11] In 2018 a second team cut the same structures in three dimensions and argued they are folds made by pressure, not by microbes. [12] That debate is over whether life goes back 3.7 billion years. It does not unsettle the 3.4 billion.

Evolution needs time, and the rocks show there has been plenty of it. Part 2 shows what happens to living things over that time.

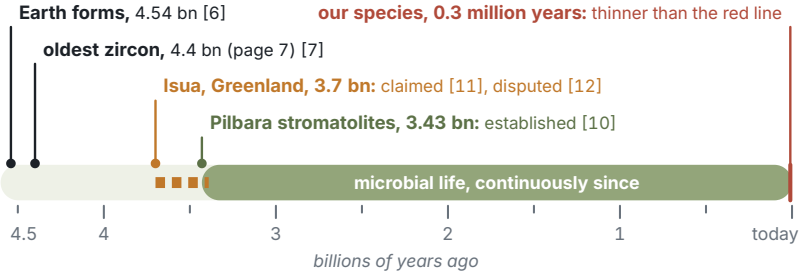


3.43 bn
years: Pilbara stromatolites

10 km
of fossil reef mapped in 2006

3.7 bn
years: the Isua claim, disputed

A stromatolite from the Dresser Formation, Pilbara, Western Australia: about 3.48 billion years old, possibly older evidence of life, but debated because little organic matter survives **[13]**. James St. John, CC BY 2.0.



CHECK IT NHM: *Origin of life on Earth* **[9]**; *Oldest evidence of life* **[13]** · Allwood 2006 **[10]**

2

PART 2

How evolution works

How do living things change over time?

Part 1 showed there has been time: life is at least 3.4 billion years old. This part shows what happens to living things over that time: natural selection watched in a real moth, the part chance plays, how one species becomes two, and a prediction about the fossil record that came true. Part 3 turns to us.

NOT THE SAME QUESTION: EVOLUTION AND THE ORIGIN OF LIFE

Evolution explains how living things change and branch once life exists. How life first arose from non-living chemistry, called *abiogenesis*, is a separate question, and still an open one. Experiments have made some of life's building blocks from simple chemicals, and many scientists favour deep-sea hot springs as the setting, but no one yet knows how the first cells formed. [9, 14] Nothing in this booklet depends on the answer: however life began, the fossils, dates and DNA show the same branching history. Further reading: page 58.

IN THIS PART

10 What makes a population change? · Variation, selection, inheritance
11 How does one species become two? · Small changes add up to new species
12 Can evolution make predictions? · A prediction that guided a search

WHAT MAKES A POPULATION CHANGE?

Variation, selection, inheritance

Evolution is change in a population's inherited features over generations. It needs three things.

Variation. Individuals differ, and some differences are inherited. They begin as chance changes in DNA (mutations), which arise in every generation. [15] A black peppered moth, from a change estimated around 1819, [16] was first recorded in Manchester in 1848. [17]

Selection. Some variants survive and breed better than others. Coal smoke blackened trees and killed pale lichens, so pale moths stood out and birds ate more of them; more black ones survived. [17]

Inheritance. Parents pass on their DNA, so survivors' features become more common. Black moths had black young; by 1895, 98% were black. [3, 17, 18]

Together, the three change a population. After the 1956 Clean Air Act cleared the soot, black moths became rare again (chart below), [19] as birds pick off moths that stand out. [20] **Populations evolve, individuals do not.** Mutation is chance; selection is not: it favours what works in today's surroundings, and never plans ahead.

A REAL CASE: THE PEPPERED MOTH IN MANCHESTER



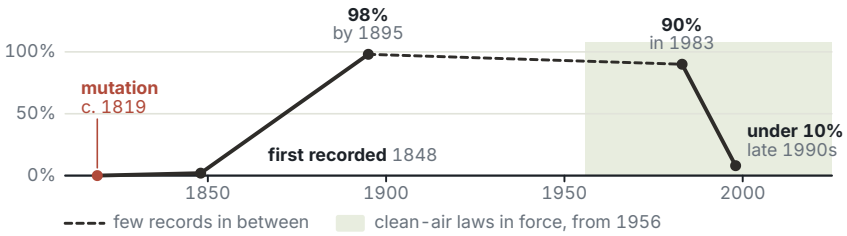
1 Variation, c. 1819
One moth is born black. It is rare.

2 Selection
Soot darkens trees; birds eat pale moths.

3 Inheritance
Black moths pass the black form on.

4 By 1895
Almost all moths are black.

SHARE OF MOTHS THAT ARE BLACK, MANCHESTER AREA



CHECK IT Van 't Hof 2016 [16] · Cook and Saccheri 2013 [17] · Cook, Dennis and Mani 1999 [19] · Cook et al. 2012 [20]

HOW DOES ONE SPECIES BECOME TWO?

Small changes add up to new species

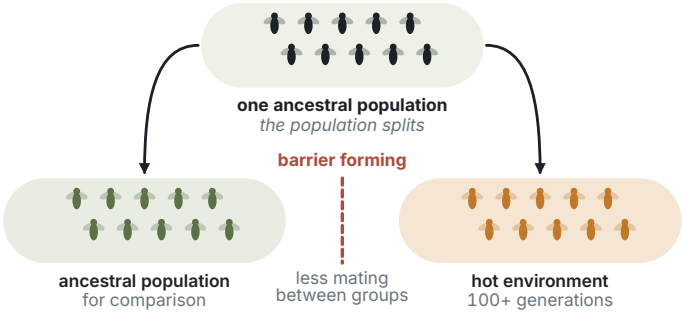
Small inherited changes add up. When two populations stop interbreeding, their changes can no longer mix, and over time they can become separate species.

Each generation differs only slightly from the last, but the differences accumulate: over thousands of generations, and the millions of years of Part 1, a population can end up very different from its ancestors. *Microevolution* is change within a population, as on the previous page; *macroevolution* is the splitting of lineages and the origin of species. They are the same process at **different scales**; no biological barrier between them is known. [18]

Chance plays a part too. Some variants spread or vanish by luck, not because they help: *genetic drift*. It is strongest in small populations, so a population that shrinks loses variation fast. [21] That is one reason people outside Africa carry less genetic diversity than Africans (pages 41–42).

New species usually form through *reproductive isolation*: populations that stop interbreeding are "on the path to becoming separate species", as the Natural History Museum puts it. [22] The process is gradual and the boundaries can blur; related species sometimes still interbreed, as our ancestors and the Neanderthals did (page 40). In the laboratory study in the diagram, fruit flies kept hot for more than 100 generations showed early signs of barriers to mating with their ancestral population: a step towards new species, **not** the whole history of one. [23]

Over millions of years, splitting populations should leave a branching tree, and a tree makes risky predictions. The next page shows one that came true.



Reproductive isolation can evolve: a step towards new species [23]

CHECK IT NHM: *What is a species?* [22] · Hsu 2024 [23] · Berkeley [18, 21]

CAN EVOLUTION MAKE PREDICTIONS?

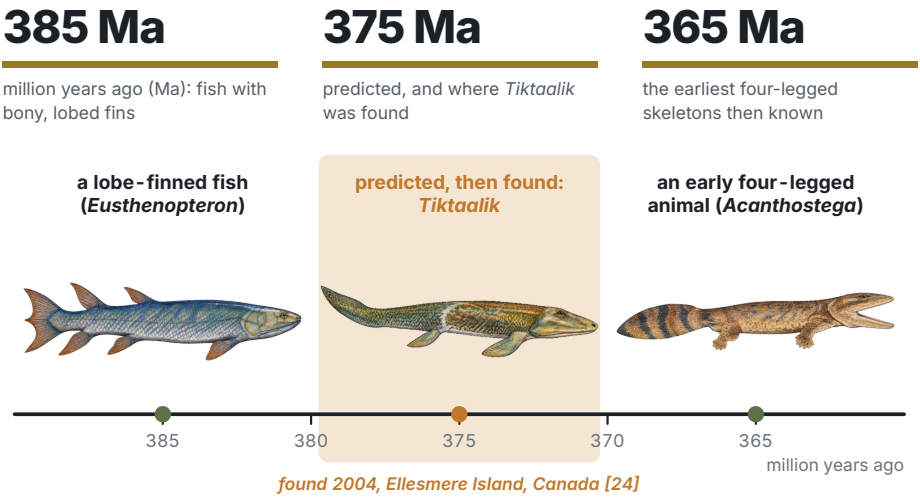
A prediction that guided a search

If fish gave rise to four-legged animals, in-between forms should lie in rocks of roughly one age. That reasoning told researchers where to look, and they found one.

By the 1990s, fish with bony, lobed fins were known from rocks about **385 million years** old, and the earliest four-legged animals from rocks about **365 million years** old. [24, 25] If one group gave rise to the other, animals part-way between them should turn up in rocks of about **375 million years**. The prediction could have failed: nothing guaranteed such fossils survived.

Neil Shubin and his colleagues looked for exposed rocks of that age, laid down in shallow streams, and found them on Ellesmere Island in Arctic Canada. In 2004, after several seasons of searching, they found *Tiktaalik*, shown in the diagram between the two. It was a fish with scales and fins, but it also had a flat head, a neck, and the bones of our arm inside its front fins: one upper bone, two forearm bones and a wrist. [24, 25]

Later finds made the timing less neat: footprints about 395 million years old, from Poland, suggest four-legged animals already existed, so *Tiktaalik* was a late survivor of an in-between group, not the ancestor itself. [26] The prediction was about where to look, and the features were as predicted. Part 3 applies the same method to us.



CHECK IT Daeschler, Shubin and Jenkins 2006 [24] · Shubin 2008 [25]

3

PART 3

Where we fit

What exactly is claimed about us, and how could it be tested?

Parts 1 and 2 gave the method and the mechanism. Now the question itself: exactly what is claimed about humans, and three tests that could prove it wrong. Then our place on the tree of life, one step closer at a time, from all life to the last two million years. Parts 4 and 5 carry out the tests.

IN THIS PART

14 What exactly is claimed about us? · Three tests of shared ancestry
15 Where do humans sit among all life? · One tree of life
16 Which groups do we belong to? · Our place among the animals
17 Who are our closest relatives? · We are apes
18 Did we come from chimpanzees? · Cousins, not ancestors
19 What does the human family look like? · A branching bush, not a ladder
20 Was there one kind of human at a time? · Several kinds of human lived at once
22 What did our relatives look like? · Different faces, overlapping times
24 What is distinctive about humans? · An unusually cultural ape

WHAT EXACTLY IS CLAIMED ABOUT US?

Three tests of shared ancestry

The claim: humans share ancestors with the other apes. If that is true, three things must be found. If any is missing, the claim is in trouble.

In full: our species arose within earlier populations, and we share ancestry with the other apes. Scientists argue over parts of the family tree because there is a real tree to investigate, not because the whole history rests on one missing skull. [27–29]

No one watched this happen, but events leave traces: a skull from Morocco, footprints in Tanzanian ash, the DNA of a Neanderthal, a join inside our second chromosome. [27, 30–34] An explanation is strong when **independent traces point the same way**, as when a court reconstructs a crime no witness saw.

Why tests? A claim that could never be wrong tells us nothing. Shared ancestry says what the evidence should look like, and each test below names what would have counted against it. Some fossils were found before the question was framed this way, but the record could still have looked different, and it does not. Each test uses a different kind of evidence, read by different people with different methods. One surprise would not overturn the picture; what matters is whether the evidence lines up overall. Parts 4 and 5 carry the tests out, each page with a **Would have challenged this if** line. First, the rest of this part shows the tree itself.

1 Older forms in older rocks

PREDICTS
relatives with mixed traits, in the right order

CHALLENGED IF
fully human skeletons in the oldest layers

EVIDENCE
Lucy, footprints, *H. erectus*, brain sizes

Part 4 · pp. 29–34

2 A branching pattern

PREDICTS
closer relatives share more of their DNA

CHALLENGED IF
a DNA tree that contradicts anatomy

EVIDENCE
genome comparisons, Neanderthal DNA

Part 5 · pp. 36, 40

3 Shared marks in our DNA

PREDICTS
damage shared with apes, and a chromosome join

CHALLENGED IF
few shared broken genes, or no join

EVIDENCE
a broken vitamin C gene, chromosome 2

Part 5 · pp. 38–39

each test could have failed; all three held

CHECK IT Smithsonian: *Human Evolution Evidence* [27] · NHM: *Modern humans* [35]

WHERE DO HUMANS SIT AMONG ALL LIFE?

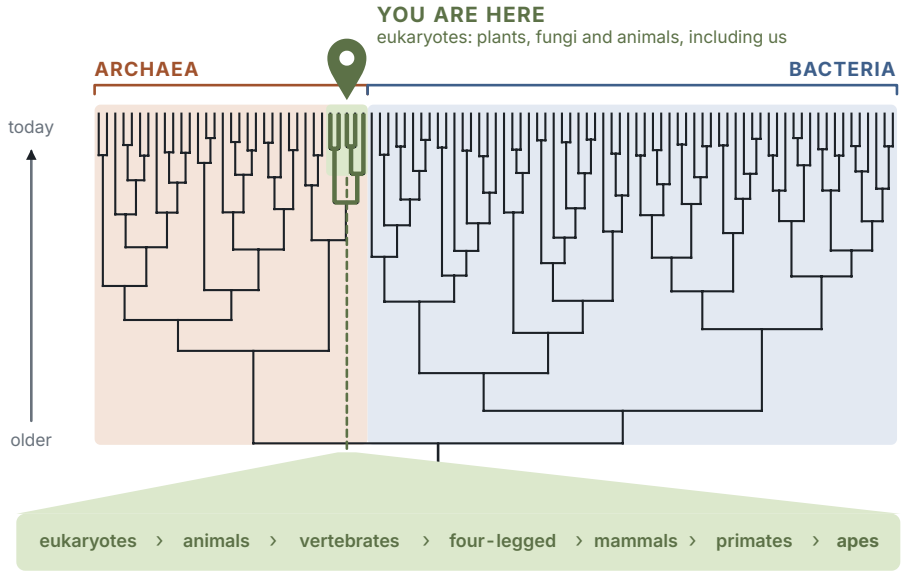
One tree of life

Every living thing belongs to one family tree. Humans are one small twig on it, in the branch called the eukaryotes.

The diagram shows the whole tree, reconstructed by comparing the DNA of thousands of living organisms. [36] Its three great branches were first recognised in 1977. Carl Woese and George Fox compared one gene that every living cell uses, and found that microbes long lumped together as "bacteria" were two separate branches of life. [37]

Most of the branches are microbes: in this reconstruction, bacteria form about two-thirds of them and archaea most of the rest. [36] Familiar plants, fungi and animals, from oak trees and mushrooms to whales and people, all sit on one small branch: the *eukaryotes*. The best current evidence places their origin among a group of archaea called the Asgard archaea, so the three-branch picture is a simplification; exactly how is still debated. [38]

The next pages zoom in along the green band, one step at a time, to our own branch.



Schematic: each tip is a large group of organisms, most of them microbes; after Hug et al. 2016 [36]. Eukaryotes grew out of the Asgard archaea; exactly where within them is still debated [38].

CHECK IT Hug et al. 2016 **[36]** · Eme et al. 2023 **[38]** · Woese and Fox 1977 **[37]**

WHICH GROUPS DO WE BELONG TO?

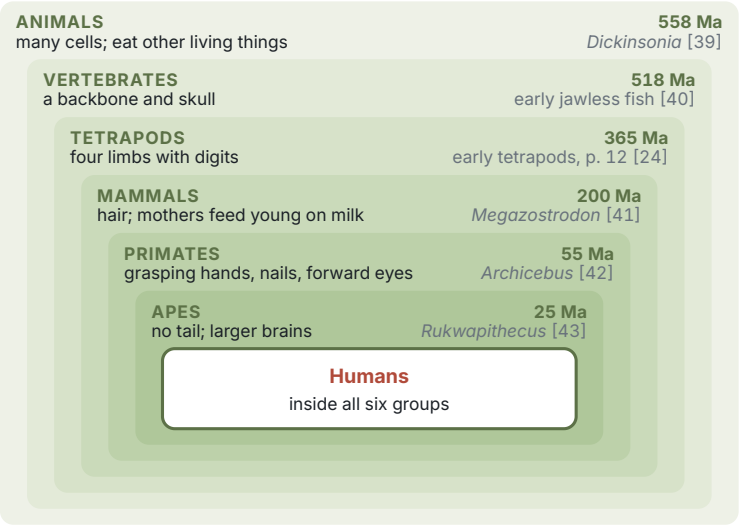
Our place among the animals

Humans belong to a series of groups, one inside the next. Each group descends from one ancestor, and each appears in older rocks than the group inside it.

We are **animals**; among them, **vertebrates**; then **tetrapods**, descended from fish like *Tiktaalik* (page 12); then **mammals**, **primates** and **apes**. The diagram gives each group's inherited feature. [24, 39–43]

Early, well-dated fossils of each group fall in the order the nesting predicts, from animals 558 million years ago to apes 25 million. [39–43] A rabbit in rock older than the first fish would break the pattern; none has been found.

The same bones, reshaped. A human arm, a whale's flipper and a bird's wing share one upper bone and two forearm bones, as do the fins of the fossil fish *Eusthenopteron* (page 12). [44] And apes, ourselves included, lost the tail other primates keep; only the tailbone, three to five fused vertebrae, remains, though a human embryo still grows a tail that then shrinks away. [45, 46] A 2024 study linked the loss to a DNA insertion in the ancestor of all apes, which apes still share. [45]



Each box sits inside the one before: every group is one branch of the group around it. Early, well-dated fossils of each group, not exact origin dates (Ma = million years ago).

CHECK IT Bobrovskiy 2018 [39] · Shu 1999 [40] · NHM [41] · Ni 2013 [42] · Stevens 2013 [43] · Daeschler 2006 [24] · Berkeley [44] · Xia 2024 [45] · Tojima 2018 [46]

WHO ARE OUR CLOSEST RELATIVES?

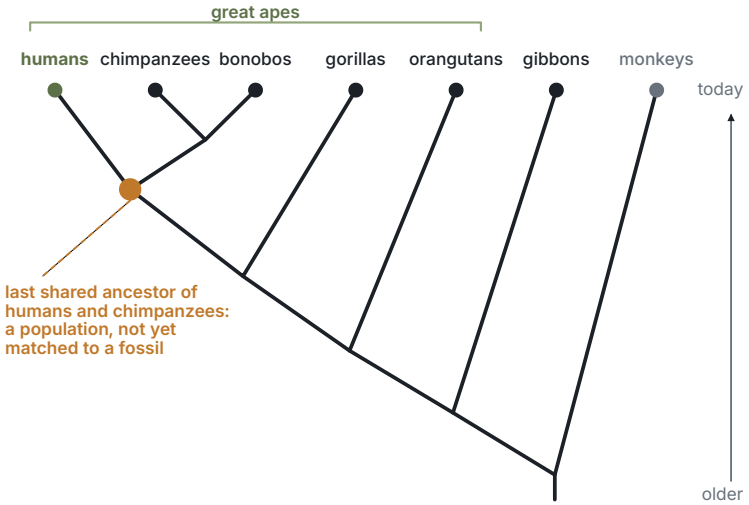
We are apes

Humans are apes. Our closest living relatives are chimpanzees and bonobos.

In the tree below, every tip is a living ape and every join is a shared ancestral population. Humans, chimpanzees and bonobos are more closely related to one another than any of them is to gorillas; we share deeper ancestry with gorillas, orangutans and gibbons, and deeper still with monkeys. [47]

It is the same kind of tree as the one on page 15, over a much shorter time: the apes split from the Old World monkeys around 25 million years ago (page 16). [43] Every living ape is a cousin branch; none is an ancestor of any other.

A prediction from geography. In each region of the world, Darwin noted in 1871, living mammals are closely related to the extinct ones there. Since our nearest relatives live in Africa, it was "somewhat more probable that our early progenitors lived on the African continent than elsewhere." [48] Early human fossils began to turn up there only in 1924, with the Taung Child; [49] today the oldest known all come from Africa (page 28).



Every living ape is a cousin branch. The tips are all alive today; the joins are shared ancestral populations.

CHECK IT Smithsonian: *Genetics* **[47]**; *Human Family Tree* **[29]** · Stevens 2013 **[43]** · Darwin 1871 **[48]**

DID WE COME FROM CHIMPANZEES?

Cousins, not ancestors

We did not descend from chimpanzees. We and they descend from one ancestral population, which lived between about 9 and 6 million years ago.

Zoom in on the join between our branch and the chimpanzees'. The 9–6 million years is an estimate from DNA: both museums quote it as what the genetic evidence shows (page 37). [35, 47] Fossils can only set a minimum: the split must be older than any fossil on our branch. **We did not descend from the chimpanzees alive today**: they have been evolving for just as long as we have. They are our cousins, not our grandparents, as the diagram below shows.

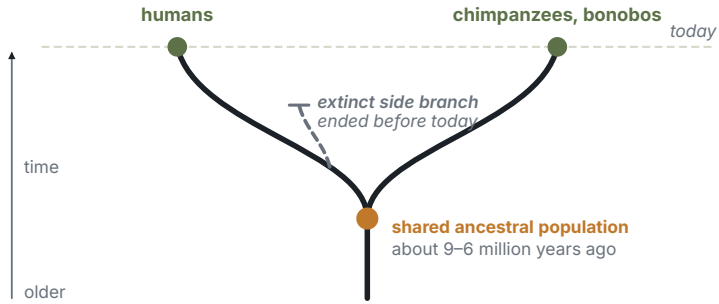
The fossil species that was the last shared ancestor **has not been identified**. Fossils near the split are informative, but calling any one of them "our proven direct ancestor" would go beyond the evidence. [28, 29] The next pages show the human side of the split: documented branches, without pretending every ancestor is known. [27, 32]

Two words help from here on. *Hominins* are the human branch after the split, including *Ardipithecus* and *Australopithecus*. The genus *Homo* is the part of that branch that includes our species and several extinct relatives. [28, 29]

9–6 million
years ago: our last shared ancestor with chimpanzees

2
closest living relatives: chimpanzees and bonobos

0
living apes that are our ancestors



cousins, not ancestors: neither living species descends from the other

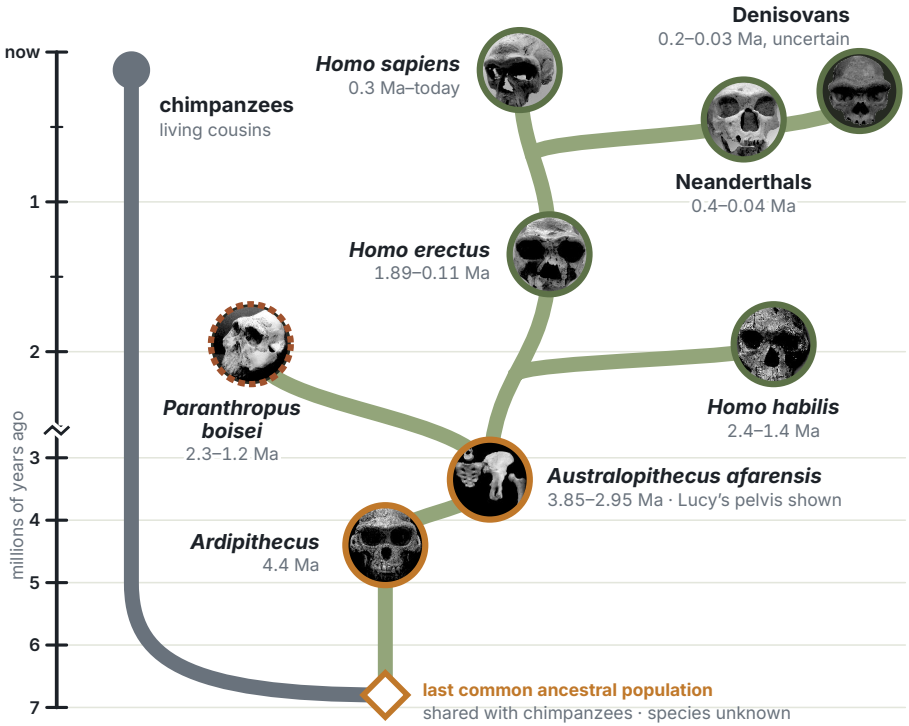
CHECK IT NHM: *Modern humans* **[35]** (genetic evidence: "between six and nine million years ago") · Smithsonian: *Genetics* **[47]** (DNA: "8 and 6 million")

WHAT DOES THE HUMAN FAMILY LOOK LIKE?

A branching bush, not a ladder

Our family tree has many branches, and most of them ended.

A **simplified** tree of selected relatives, below: oldest at the bottom, today at the top. Each circle shows a real fossil of that species (Lucy's pelvis for hers; the Harbin skull, identified in 2025, for the Denisovans) [50]. A name on a branch means "a documented member of this part of the tree", **not** "proved parent of the next species"; joining lines are proposed relationships. Even Lucy's species may be a side branch (page 29). [28, 29, 51]



KEY *Time axis: the last 2.5 million years are stretched (break mark on the axis)*

- early hominins + Australopithecus
- Paranthropus · robust-jawed side branch
- Homo · our genus
- simplified possible relationship
- chimpanzee lineage
- last common ancestral population

Not shown: Sahelanthropus · Orrorin · other Australopithecus · P. robustus · H. heidelbergensis · H. naledi · H. floresiensis (see page 20)

CHECK IT Ranges: Smithsonian species pages [28] · NHM 2025 [51] · Tree: George Philippou

WAS THERE ONE KIND OF HUMAN AT A TIME?

Several kinds of human lived at once

For most of the last two million years, several kinds of human were alive at the same time.

Each bar is one species' known time range, from its oldest to its youngest dated fossil. [28] Dashed bars are uncertain or inferred, and the chart continues on the facing page.

Look along the orange band: on the Smithsonian's dates, at least five kinds of human were alive 250,000 years ago, from *H. naledi* in southern Africa to early *H. sapiens*. [28, 52]

A branch ending means a lineage did not survive as a separate species; its genes may live on in us (page 40). [32, 53]



HOW TO READ THE CHART

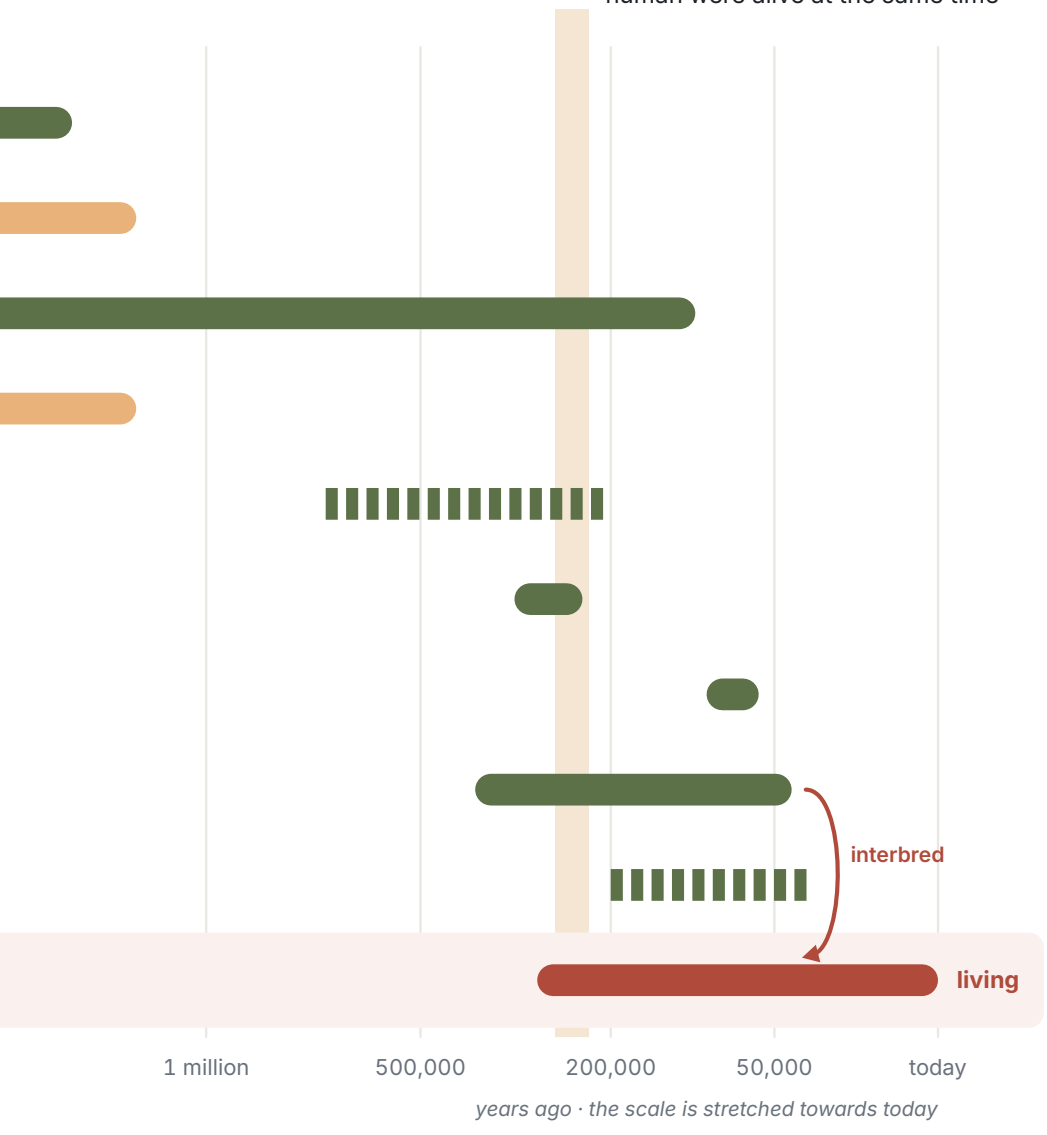
- time range known from fossils
- range uncertain, debated or inferred
- *Paranthropus*: a side branch
- our species

Each tile shows a real skull of that species; pages 22–23 show four of them larger.

CHECK IT Smithsonian species pages [28] · NHM: *Homo naledi* [52]; *Denisovans* [54]

250,000

years ago: at least five kinds of human were alive at the same time



WHAT DID OUR RELATIVES LOOK LIKE?

Different faces, overlapping times

Each relative had its own mix of features. None is simply a smaller version of us.

Homo habilis and *Paranthropus boisei* lived in eastern Africa at the same time, about 2.3 to 1.4 million years ago. *H. habilis* had a small braincase and a lighter face than *P. boisei*; it is linked with early stone tools, though other hominins may have made the first ones (page 31). *P. boisei* had huge cheek teeth, on a side branch that left no descendants. [28, 55]



***Homo habilis*, KNM-ER 1813**

2.4–1.4 million years

Look for: a small braincase, about 510 cm³, but a lighter face and smaller teeth than *Paranthropus*.

[28]

José-Manuel Benito Álvarez, public domain



***Paranthropus boisei* (cast)**

2.3–1.2 million years

Look for: huge cheekbones and a bony crest on top of the skull for massive chewing muscles: "Nutcracker Man".

[28]

Daderot, CC0

CHECK IT Smithsonian species pages [28]; KNM-ER 1813 [49] · NHM: *Homo naledi* [52] · Berger et al. 2015 [56]

Small-brained, and late. *Homo floresiensis* lived on Flores, Indonesia: its bones are 100,000–60,000 years old, its stone tools as young as 50,000. [28, 57] *Homo naledi*, from the Rising Star cave in South Africa, is known from at least 15 individuals in one cave system. [52, 56] Neither fits a ladder from small brains to large; both fit a branching tree (page 33). Part 4 tests this tree against the rocks.



***Homo floresiensis*, Flores**

bones 100,000–60,000 years

Look for: a tiny braincase, about 400 cm³, on an adult about 1.1 m tall.

[28]

Rama, CC BY-SA 3.0 fr



***Homo naledi*, DH1**

335,000–236,000 years

Look for: a small braincase, about 465–560 cm³, with small, human-like teeth, yet it lived alongside early *H. sapiens*.

[28, 52, 56]

Berger et al. 2015, CC BY 4.0

WHAT IS DISTINCTIVE ABOUT HUMANS?

An unusually cultural ape

Every species is a product of evolution, and every one is unique. What is distinctive about humans is not one trait but a package that reinforces itself, with culture that builds on itself at its centre. Many animals have pieces of it; humans combine them to an unusual degree.

Culture that accumulates. We keep what earlier generations learned and improve on it. [58]

Open-ended language. Human language can build an unlimited range of expressions from a finite set of elements. [59]

Cooperation with strangers. Experiments find a readiness to cooperate with unrelated people with no close parallel in other animals. [60]

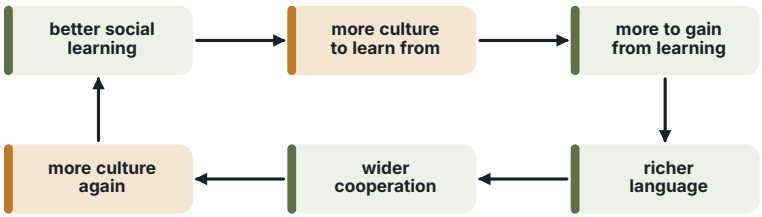
Social minds. Children aged two and a half matched chimpanzees on physical skills but beat them on social ones, such as learning from others. [61]

A long childhood. We mature late, and women live long after menopause, which one hypothesis links to grandmothers feeding grandchildren. [62]

Hands and feet. Walking upright (page 30) leaves the hands free, and human hands have strong pinch grips suited to making stone tools (page 31). [63]

A flexible range. One African lineage spread into nearly every land habitat, from deserts to the Arctic (page 41). [64]

The traits feed one another (loop below), and culture has shaped our genes: the ability to digest milk as an adult spread among cattle herders. [65, 66] The loop is a model, not a law. It shows why "the biggest brain" is not the answer: what sets us apart is knowledge that accumulates. Next, the rocks (Part 4).



a proposed loop: culture and genes shape each other [66]

CHECK IT Henrich 2016 [58] · Laland et al. 2010 [66]

4

PART 4

Test 1: the rocks

Are older forms found in older rocks?

Test 1 of the three set on page 14: if we evolved, older rocks should hold older relatives, less like us, in the right order. First, how rock ages are measured and where the fossils come from; then the finds, oldest first: an upright ape, footprints, stone tools, a body like ours, growing brains and the first of our own species. Part 5 checks the same history in DNA.

IN THIS PART

26 How are fossils dated? · Ages are dated, not guessed
27 Are the fossils real? · Real specimens with catalogue numbers
28 Where were they found? · Every find has an address
29 When did walking upright begin? · Lucy's species walked upright
30 Is there a direct trace of it? · Footprints 3.6 million years old
31 When did tools appear? · Stone tools, in order of age
32 When did a body like ours appear? · *Homo erectus*: a body built like ours
33 Did brains grow steadily? · Brains grew, but not in a straight line
34 Where did our species arise? · Our species arose in Africa

HOW ARE FOSSILS DATED?

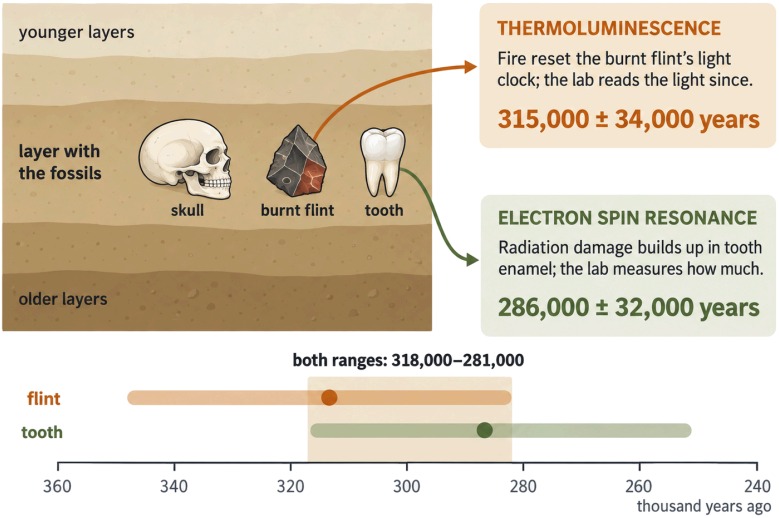
Ages are dated, not guessed

Fossil ages in this booklet come from dating the rocks and materials around them, with stated uncertainties. At Jebel Irhoud, two methods on two materials agree.

Rock layers give a sequence: an undisturbed lower layer is older than the one above. Volcanic minerals, burnt flint and tooth enamel can be dated by physical methods, and scientists compare methods, context and uncertainties. **Carbon-14 is not used to date million-year-old fossils**; it only works to about 50,000 years. [67]

The diagram shows how this worked at Jebel Irhoud, Morocco, where the oldest known fossils of our species were found (page 34). Thermoluminescence dated flint tools burnt in ancient fires and buried with the remains: **315,000 ± 34,000 years**. A second method, electron spin resonance on a human tooth from the site, gave 286,000 ± 32,000 years: a different material and physical principle, overlapping within their uncertainties. [31, 68]

Dates are estimates with stated limits, cross-checked by two methods that share one assumption, the site's radiation history.



Two methods, two materials, one layer: the two ranges overlap.

CHECK IT NHM: *Jebel Irhoud* [68] · Smithsonian: *Dating* [67] · Richter 2017 [31]

ARE THE FOSSILS REAL?

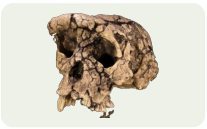
Real specimens with catalogue numbers

The fossils on these cards are real objects with catalogue numbers you can look up.

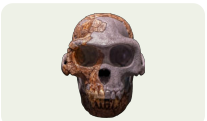
These are **individual fossils**, not drawings of imagined ancestors. Each card gives a catalogue number, a find site, an age and where to check it. [49, 69] Whether Toumaï's species walked upright is still argued; a 2026 study adds evidence that it did. [70–72]



millions of years ago (Ma) · stretched scale, so recent finds stay visible



1 TM 266-01-060-1 "Toumaï"
Sahelanthropus tchadensis · skull
Toros-Menalla, Chad — **about 7–6 million years**
Look it up: Smithsonian catalogue [49]



2 ARA-VP-6/500 "Ardi"
Ardipithecus ramidus · skull (cast shown)
Aramis, Ethiopia — **about 4.4 million years**
Look it up: Smithsonian catalogue [49]



3 AL 288-1 "Lucy"
Australopithecus afarensis · pelvis and thigh bone shown
Hadar, Ethiopia — **about 3.2 million years**
Look it up: Smithsonian [49]; NHM [73]



4 KNM-ER 3733
Homo erectus (some say *H. ergaster*) · skull (replica shown)
Koobi Fora, Kenya — **about 1.6 million years**
Look it up: Smithsonian catalogue [49]; Lepre and Kent 2015 [74]



5 La Chapelle-aux-Saints 1
Neanderthal · skull
La Chapelle-aux-Saints, France — **about 60,000 years**
Look it up: Smithsonian catalogue [49]

None of these is proven to be anyone's direct ancestor. The Natural History Museum, London, displays Gibraltar 1, the first adult Neanderthal skull ever found (1848). [69]

CHECK IT Smithsonian fossil catalogue [49] · NHM: *First adult Neanderthal skull* [69]

WHERE WERE THEY FOUND?

Every find has an address

Every find has an address: a site, a rock layer and a measured age.

A fossil is not simply "an old bone". It has a find site, a geological setting, a specimen number and, for well-documented finds, a dated range. The map shows where the finds in this booklet were made, across millions of years and a vast area of Africa, Europe and south-west Asia. [49]

The map locates **discoveries**, not a path walked by anyone. Fossils are recovered only where bodies were preserved and the right deposits are exposed and searched; a blank area is not evidence that no one lived there. The rest of this part looks at what the finds show, oldest first.



CHECK IT Sites and ages: [30, 31, 34, 70, 75, 76] · NHM: *Homo erectus* [77] (Dmanisi 1.8–1.85 million years) · Ferring 2011 [76]

WHEN DID WALKING UPRIGHT BEGIN?

Lucy's species walked upright

3.2 million years ago, Lucy's species walked upright, with a brain about a third the size of ours.

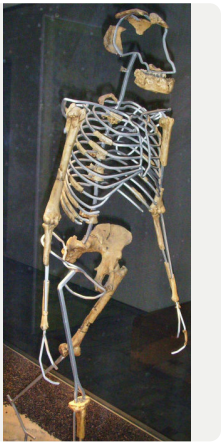
THE FIRST TEST: OLDER FORMS IN OLDER ROCKS · SEE PAGE 14

WOULD HAVE CHALLENGED THIS IF fully human skeletons turned up in the oldest rocks.

Early human fossils show mixtures of ape-like and human-like traits, and *Australopithecus afarensis*, from eastern African rocks **3.85 to 2.95 million years old**, shows this especially well. [28, 73]

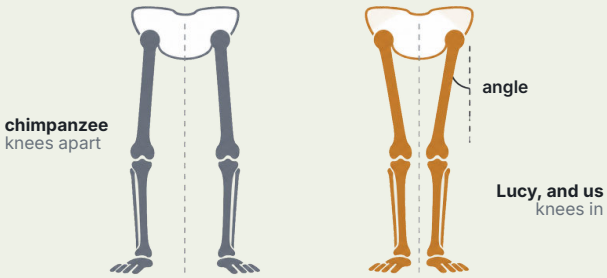
Its best-known skeleton, Lucy (below), was found at Hadar in 1974 and named that night after a Beatles song; in Ethiopia she is *Dinknesh*, "you are marvelous". [49] Her pelvis, thigh bone and spine show a body built to walk on two legs. The key clue is the **angle of her thigh bone**: it slopes inwards from hip to knee, as ours does, putting her weight over her feet. A chimpanzee's runs straight down, with the knees apart. A foot bone of her species is arched like ours; long arms and curved fingers suggest it still climbed. [49, 73, 78] The diagram compares her thigh bones with a chimpanzee's (**observed**) and lists what that suggests (**inferred**).

A 2025 study suggests Lucy's species may be a side branch, not a direct ancestor. That would not undermine the tree. [51]



Lucy (AL 288-1) mounted for display, Senckenberg Museum. Chiswick Chap, CC BY-SA 3.0.

OBSERVED the thigh-bone angle



INFERRED what the shapes imply

- Angled thigh bones put knees and feet under the body.
- Long arms and curved fingers: she may have climbed trees.
- Behaviour needs care: the same bones can be read two ways.

CHECK IT NHM: *Lucy's species* [73] · Smithsonian: *Species* [28]

IS THERE A DIRECT TRACE OF IT?

Footprints 3.6 million years old

Footprints in volcanic ash show upright walking 3.6 million years ago, long before our species existed.

THE FIRST TEST: OLDER FORMS IN OLDER ROCKS · SEE PAGE 14

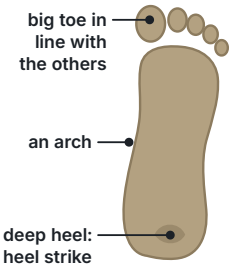
WOULD HAVE CHALLENGED THIS IF modern-looking feet had appeared with no earlier mixed forms.

In 1978, at Laetoli in Tanzania, Paul Abell of Mary Leakey's team found a 27-metre trail of footprints (photo, below left), left by upright walkers crossing wet volcanic ash **3.6–3.7 million years ago**. [30] The foot diagram marks three features of the tracks: a deep heel strike, an arch and a big toe in line with the others. Together they show upright walking like ours. [30, 73, 79] The strip below shows how they survived: the ash set, then new ash buried it.

Fossils of Lucy's species occur in the same layers, making it the likely track-maker. Footprints carry no name tag: the manner of walking is clearer than the species. [30] Bones show what early humans were built for; these tracks show what they did. [73]

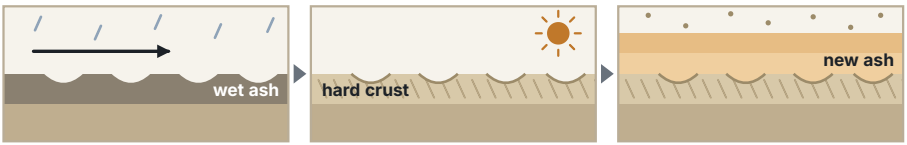


Replica of the Laetoli trackway: two upright walkers crossed wet ash about 3.6 million years ago [30]. Momotarou2012, CC BY-SA 3.0.



one print, schematic

HOW THE PRINTS SURVIVED



1 Walkers cross fresh, wet ash
2 The ash sets hard
3 New ash buries the prints

CHECK IT Smithsonian: *Laetoli* [30] (3.6 million) · NHM: *Lucy's species* [73] ("nearly 3.7 million")

WHEN DID TOOLS APPEAR?

Stone tools, in order of age

Stone tools are widespread by about 2.6 million years ago; possible older ones are debated. More complex kinds appear in later layers.

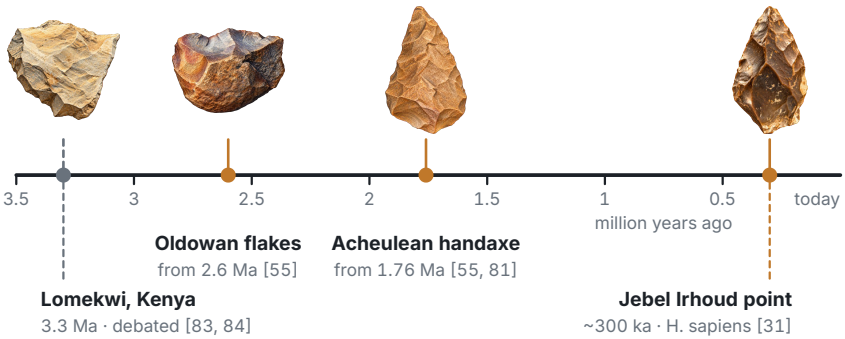
Tools are a second kind of trace. Stone survives far better than bone, and tools are found in dated layers, so the way they changed can be read in order. [55]

Deliberately flaked tools, with sharp flakes struck from stone cores, appear in eastern Africa by about **2.6 million years ago**. [80] By about **1.76 million years ago**, toolmakers were shaping large, symmetrical handaxes. [55, 81] At Nyayanga, Kenya, similar flakes 2.6–3.0 million years old lie beside teeth of *Paranthropus*, so the first toolmakers may not all have been *Homo*. [82] The burnt flints dated at Jebel Irhoud (page 26), about 300,000 years old, are carefully shaped tools made by early members of our own species. [31]

One older claim is still debated. In 2015 a team reported cruder tools at Lomekwi, Kenya, dated to **3.3 million years**, older than any known member of *Homo*. [83] Other researchers have questioned whether the tools really came from the dated layer. [84] As with the oldest signs of life (page 8), the claim is published, challenged and checked in the open.

The timeline below shows dated examples of each stone-tool tradition (Lomekwi, debated, is the oldest claim): simple flakes first, shaped handaxes later, finer points later still. Traditions overlapped in time and place.

DATED EXAMPLES OF STONE-TOOL TRADITIONS



CHECK IT Smithsonian: *Stone tools* [55] · Lepre 2011 [81] · Harmand 2015 [83] · Archer 2020 [84]

WHEN DID A BODY LIKE OURS APPEAR?

Homo erectus: a body built like ours

About 1.9 million years ago a new kind of human appears: taller, with long legs and shorter arms, and the first to spread beyond Africa.

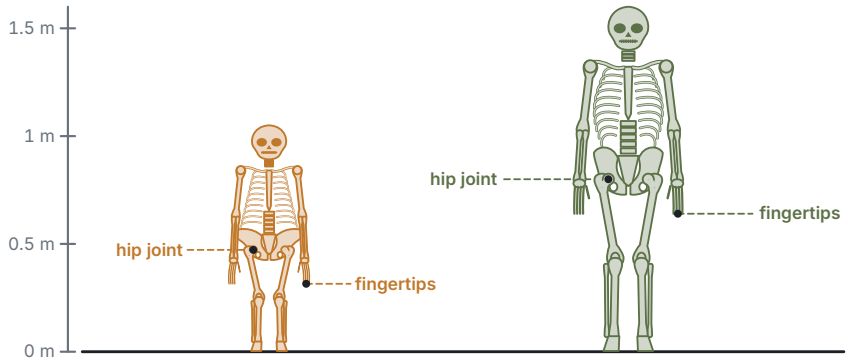
THE FIRST TEST: OLDER FORMS IN OLDER ROCKS · SEE PAGE 14

WOULD HAVE CHALLENGED THIS IF a fully human body build were older than upright walking.

Homo erectus lived from about **1.89 million to 110,000 years ago**. [28, 85] Its best-known skeleton, the Turkana Boy, found by Kamoya Kimeu at Nariokotome, Kenya, in 1984, is about **1.6 million years old** [86] and 40% complete. He died aged eight or nine, already 1.6 metres tall, with a brain-case of 880 cm^3. [49]

Lucy's species had long arms and short legs, still suited to climbing. *H. erectus* is the oldest known species with our proportions: long legs and shorter arms, built for walking, and possibly running, long distances. [28] The diagram draws both at the same scale.

It was also the first human species known outside Africa. Fossils at Dmanisi, Georgia, are about 1.8 million years old (page 28), and the species later reached China and Indonesia. [28, 77] The first handaxes appear in the same period (page 31). [81]



Lucy's species, 3.2 Ma
1.05 m · long arms, short legs
fingertips reach towards the knee

Turkana Boy, *H. erectus*, 1.6 Ma
1.6 m at age 8–9 · long legs,
shorter arms: a body like ours

anatomy simplified; heights and limb proportions to scale [28, 49, 73]

CHECK IT Smithsonian: *Homo erectus* [28]; KNM-WT 15000 [49] · NHM: *Homo erectus* [77]

DID BRAINS GROW STEADILY?

Brains grew, but not in a straight line

Over four million years, braincases roughly quadrupled in size. Two late species stayed small-brained, as a branching tree allows.

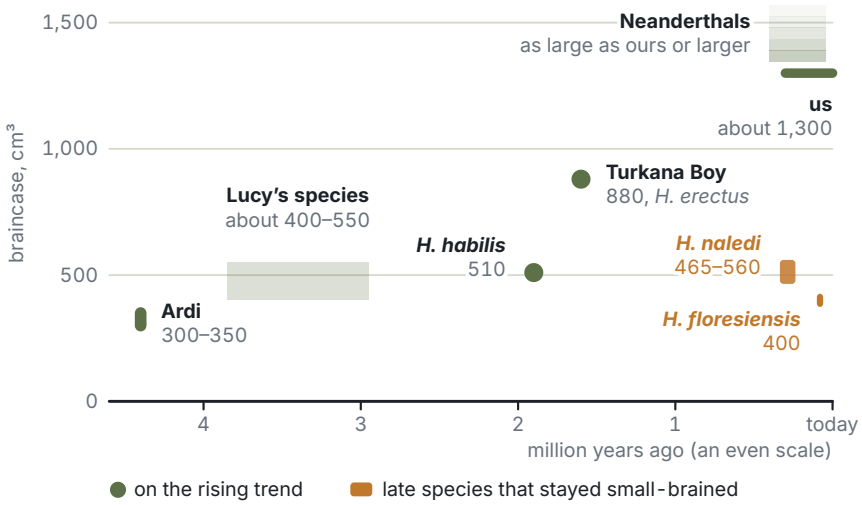
THE FIRST TEST: OLDER FORMS IN OLDER ROCKS · SEE PAGE 14

WOULD HAVE CHALLENGED THIS IF modern-sized braincases were found among the oldest hominins.

The chart plots selected specimens and species that have a published braincase volume, in cubic centimetres (cm^3), against age, on an even time scale. It is a selection, not a complete series. [28, 49]

The rise was not steady. For more than two million years braincases stayed small, from Ardi to *Homo habilis*; then they grew, fastest between about **800,000 and 200,000 years ago**. [28, 87, 88] Neanderthal brains were as large as ours, at the large end of our range. [28, 89]

The exceptions matter too: *H. naledi* and *H. floresiensis* stayed small-brained long after bigger-brained humans appeared. [56, 57, 90] A ladder cannot explain that; a branching tree, whose branches change in different ways, can.



CHECK IT Smithsonian species pages [28], specimen pages [49] and *Brains* [88] · VanSickle 2020 [89] · Berger et al. 2015 [56]

WHERE DID OUR SPECIES ARISE?

Our species arose in Africa

The oldest fossils of our species, about 300,000 years old, come from Africa.

THE FIRST TEST: OLDER FORMS IN OLDER ROCKS · SEE PAGE 14

WOULD HAVE CHALLENGED THIS IF reliably dated *Homo sapiens* fossils outside Africa were older than any African ones.

At Jebel Irhoud, Morocco, fossils with a mix of modern and older features are dated to about **300,000 years ago** (page 26). [34, 35] Others, mapped below, come from Omo in Ethiopia, where the Omo I skull lies under volcanic ash dated to at least 233,000 years. [75] A partial skull from Florisbad, South Africa, may be an early member of our species or a close relative; its 1996 date of about 260,000 years is questioned. [35, 91]

No one has found "the first human": species arise as populations change, and finds from Morocco to South Africa suggest ours took shape across the continent. [35, 64, 92] The one claim from outside Africa, Apidima 1 in Greece (over 210,000 years old; a distorted skull with no archaeological context), is still younger than Jebel Irhoud. [93] The fossil evidence points clearly one way: **our species evolved in Africa.** Part 5 checks it against our DNA.





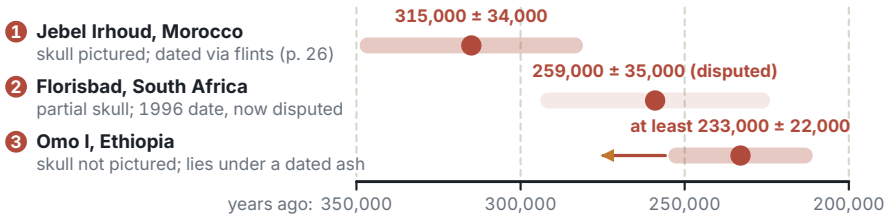
Jebel Irhoud 1 (cast)

a face much like ours on a longer braincase [31, 34]

Found by miners in **1961** [94]; first thought about 40,000 years old [68]. Site redated in **2017** to about 315,000 years, from flints [31], not this skull.

Jonathan Chen, CC BY-SA 4.0

HOW OLD: EACH AGE WITH ITS UNCERTAINTY



CHECK IT NHM: *Modern humans* [35]; *Jebel Irhoud* [68] · Smithsonian: *Homo sapiens* [95] · Gibbons 2017 [94]

5

PART 5

Tests 2 and 3: our DNA

Does our DNA tell the same story as the rocks?

Tests 2 and 3: DNA is a second record, read in laboratories rather than dug from the ground. If it told a different story from the rocks, the case would fail. First the two tests, closer relatives sharing more DNA and the same scars in the same places; then what DNA adds: Neanderthals, leaving Africa, and whether we ever descended from just two people. Part 6 weighs it all.

IN THIS PART

36 Does DNA show a family tree? · The closer the relative, the closer the DNA
37 Can DNA date the split? · A second clock, in our DNA
38 What do shared mistakes show? · The same scars in the same places
39 Why do we have 23 pairs of chromosomes? · Two ape chromosomes, joined
40 Did we meet the Neanderthals? · Neanderthal DNA lives on in us
41 How did people leave Africa? · In stages, meeting others on the way
42 How big were our ancestral populations? · A history written in populations
43 Was there a first human couple? · A common ancestor is not a first couple

DOES DNA SHOW A FAMILY TREE?

The closer the relative, the closer the DNA

The more closely related two species are, the more alike their DNA: the pattern a family tree predicts.

THE SECOND TEST: A BRANCHING PATTERN · SEE PAGE 14

WOULD HAVE CHALLENGED THIS IF the DNA tree consistently contradicted the anatomical tree.

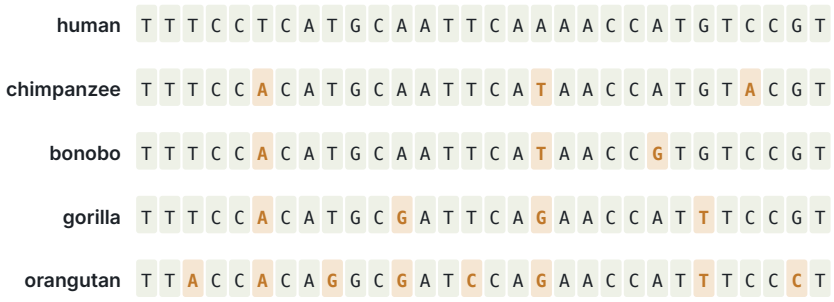
A *genome* is an organism's complete set of DNA. Human and chimpanzee genomes line up along almost their whole length, with differences built up since the lineages split. [96]

What matters is the **pattern**: in the illustrative rows below, the more distant the relative, the more letters differ from ours. The real tree is inferred from many regions of the genome, and single genes can differ.

How similar we are depends on what is counted. Where the DNA lines up, about **1.2%** of letters differ; counting insertions and deletions brings the difference to about **5%**. That does not make us "98.8% identical": complete ape genomes, published in 2025, show larger differences in regions that are hard to line up. [47, 97]

THE SAME SCIENCE, IN DAILY USE

Paternity tests and court cases rely on the same reasoning: more shared DNA means a closer relationship. Applied to species, it gives the tree.



illustration, not real sequence data: letters that differ from the human row are shaded.

Closer relatives share more: the pattern a branching tree predicts.

CHECK IT Smithsonian: *Genetics* [47] · Chimpanzee Consortium 2005 [96] · Yoo 2025 [97]

CAN DNA DATE THE SPLIT?

A second clock, in our DNA

DNA differences build up over time, so they give a second clock for when two species split. Estimates for us and chimpanzees vary with the model and the mutation rate.

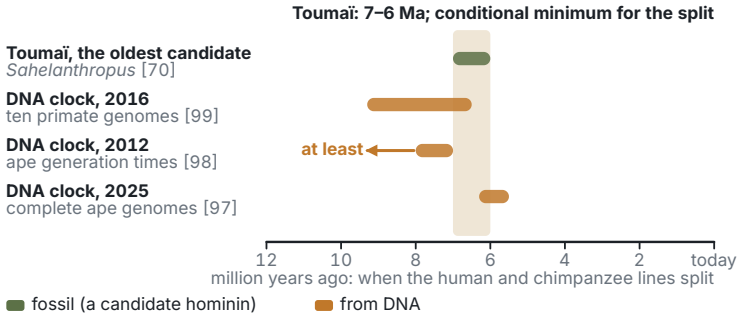
Every generation adds a few new DNA changes; once two populations stop interbreeding, their changes pile up separately. Roughly, the differences between two genomes, divided by how fast changes arise, give the time since they split: a *molecular clock*. [98, 99]

The clock's speed varies, so it is measured in living apes first. A 2016 study of ten primates dates the average human–chimpanzee DNA difference to about **12 million years** ago. The split is younger, **6.5–9.3 million years ago**, because much of that variation already existed in the ancestral population before it divided. A 2012 study, using the ages at which wild apes have young, gives **7–8 million years or more**. [98, 99] A 2025 study of complete ape genomes, with a different model, gives 5.5–6.3 million years. [97] All three use mutation rates and generation times, not a fossil date for the split.

Fossils give a check of a different kind: if *Toumaï* (page 27), about 7–6 million years old, is a hominin, the split is at least that old. That fits the earlier DNA estimates better than the 2025 one, as the chart shows. [70] The museums' 9–6 million (page 18) is quoted from DNA too. [35, 47]

THE SAME SCIENCE, IN DAILY USE

Counting changes in HIV genomes traced the pandemic back to Kinshasa in the 1920s, decades before the virus was identified. [100]



CHECK IT Moorjani et al. 2016 [99] · Langergraber et al. 2012 [98] · NHM: *Modern humans* [35] · Smithsonian: *Genetics* [47] · Faria 2014 [100]

WHAT DO SHARED MISTAKES SHOW?

The same scars in the same places

Humans and other apes carry the same broken genes in the same places: some of the clearest DNA evidence is in what is broken.

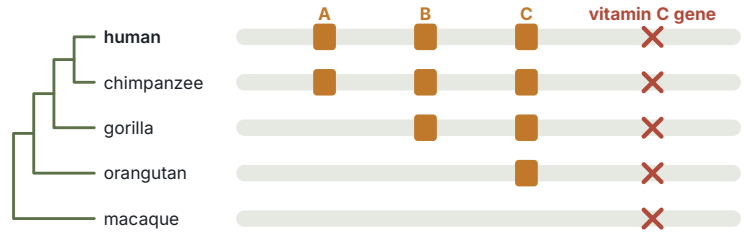
THE THIRD TEST: SHARED MARKS IN OUR DNA · SEE PAGE 14

WOULD HAVE CHALLENGED THIS IF humans shared almost none of their broken genes or virus insertions with other apes.

A broken vitamin C gene. Most mammals make their own vitamin C. Humans cannot, so we must eat it or get scurvy. We still carry the gene for the last step in making it, but it is broken. [101] Chimpanzees, gorillas, orangutans and macaques carry a broken copy too. [102, 103] In one stretch compared in humans, chimpanzees, orangutans and macaques, all four are missing the same letter that a working rat gene has. [102] The simplest explanation: it broke once, in a common ancestor, and its descendants inherited the damage.

Old viruses in our DNA. Some viruses copy themselves into the DNA of cells they infect; in an egg or sperm cell, the copy is inherited. About 8% of the human genome is made of such remnants. [104] Almost all of the insertions found in both species sit at matching places; only a small number are found in one alone. [96] Some are shared with gorillas and orangutans too, in a nested pattern that matches the family tree. [105]

Insertion sites are not entirely random, but they are spread over millions of positions, so many matching insertions with a nested pattern across species are far better explained by inheritance than by coincidence. The diagram shows the nested pattern.



CHECK IT Nishikimi 1994 **[101]** · Ohta and Nishikimi 1999 **[102]** · Mansueto 2024 **[103]** · Human Genome Consortium 2001 **[104]** · Chimpanzee Consortium 2005 **[96]**: peer-reviewed; no museum page

WHY DO WE HAVE 23 PAIRS OF CHROMOSOMES?

Two ape chromosomes, joined

Our chromosome 2 is two ape chromosomes joined end to end, and the join can still be seen.

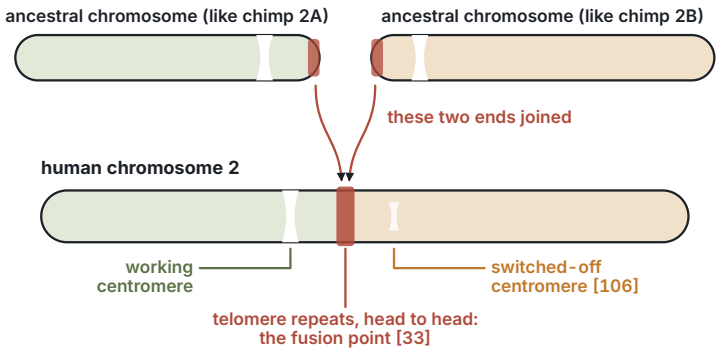
THE THIRD TEST: SHARED MARKS IN OUR DNA · SEE PAGE 14

WOULD HAVE CHALLENGED THIS IF chromosome 2 matched one ape chromosome, with no sign of a join.

Humans have **23 pairs** of chromosomes. Chimpanzees, gorillas and orangutans have **24**. If we share ancestors with them, one of our chromosomes should be two of theirs joined together. If no such join existed, the idea would be in trouble. [33, 106, 107]

Two clues show the join. **The ends.** Telomeres, a distinctive repeated sequence, normally cap the ends of chromosomes. Human chromosome 2 matches two separate ape chromosomes (the diagram's 2A and 2B), and in 1991 researchers found head-to-head telomere repeats **at the junction between the two matching halves**, exactly where two ends would have met. [33, 107] **The waist.** Every chromosome has one centromere, the pinched point the cell grips when it divides. Chromosome 2 has one working centromere and the switched-off remains of a second, from the other ancestral chromosome. [106, 108] The diagram shows both.

The join does not say when the fusion happened. But it is the physical scar that shared ancestry predicted, found where it should be. That completes the three tests.



drawn to scale: positions from the human genome sequence

CHECK IT IJdo 1991 **[33]** · NHGRI 2005 **[106]** · Avarello 1992 **[108]**: no museum page

DID WE MEET THE NEANDERTHALS?

Neanderthal DNA lives on in us

Living people with ancestry outside sub-Saharan Africa carry about 2% Neanderthal DNA. Our ancestors met them and had children.

THE SECOND TEST: A BRANCHING PATTERN · SEE PAGE 14

WOULD HAVE CHALLENGED THIS IF Neanderthal DNA were no closer to ours than an ape's.

DNA sequenced from Neanderthal bones shows that Neanderthals and the ancestors of many living people *interbred*. [32, 109] One 2025 study of early modern human genomes in Europe dates the main mixing to about **45,000–49,000 years ago**. [110, 111]

Denisovans, a sister group of the Neanderthals, also left DNA in living people: about **4–5%** in Papuans and some other populations of Island Southeast Asia and Oceania, and more in the Philippine Ayta, who carry the most Denisovan ancestry yet found (about 30–40% more than Papuans). [53, 54, 112, 113]

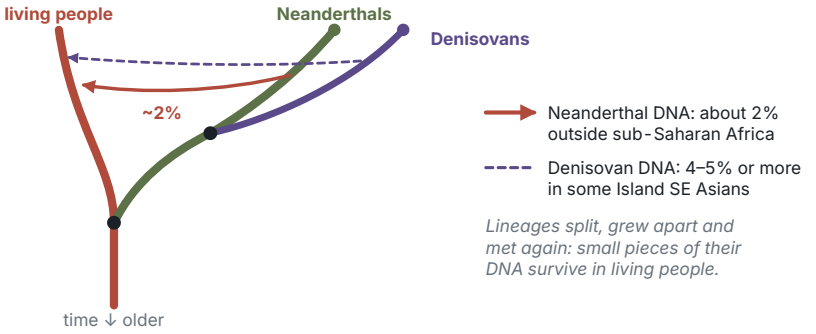
Lineages split, developed separately and sometimes met again, as the diagram shows.

At the Natural History Museum. Ancient DNA can even read appearance: the genome of Cheddar Man, a 10,000-year-old Briton in the Museum's Human Evolution gallery, points to likely dark skin and pale eyes. [114]

~2%
Neanderthal DNA outside sub-Saharan Africa

45–49 ka
thousand years ago (ka): main Neanderthal mixing

4–5%+
Denisovan DNA in some Island SE Asians



CHECK IT NHM: *Neanderthals* **[109]**; *2024 news* **[111]** · Smithsonian: *Ancient DNA* **[112]**

HOW DID PEOPLE LEAVE AFRICA?

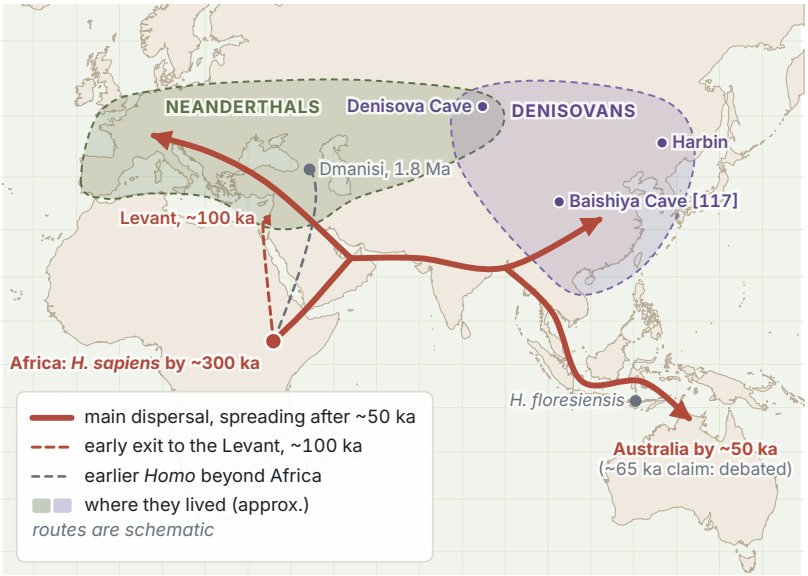
In stages, meeting others on the way

Our species spread out of Africa in stages, and met other kinds of human on the way.

Our species arose in Africa and later spread worldwide. African populations still hold the greatest genetic diversity of any people today, as expected where a species has lived longest. [64] Some groups left early, reaching the Levant by about **100,000 years ago**, but most ancestry of living people outside Africa traces to a later dispersal. [35, 64] Its members had left Africa and were still one population when they mixed with Neanderthals, about 45,000–49,000 years ago; after that, they spread across Eurasia. [110, 111]

People reached Australia by about **50,000 years ago**. One site's date of about 65,000 years is still debated. [115, 116] On the way, our ancestors met other human lineages: Neanderthals, from Europe to Central Asia, and Denisovans further east, from the Altai to the Tibetan Plateau and northern China. Shading shows roughly where each lived. [32, 50, 53, 117]

No neat arrow tracks every journey. The map shows broad regions and marks routes as uncertain. [64] The Americas, reached later from Siberia, are off the map. [118]



CHECK IT NHM: *2024 news* [111] · Sümer 2025 [110] · O'Connell 2018 [116] · Clarkson 2017 [115]

HOW BIG WERE OUR ANCESTRAL POPULATIONS?

A history written in populations

Our DNA records how ancestral populations shrank and grew over hundreds of thousands of years.

Every person inherits DNA from many ancestors, and different stretches of a genome have different histories. From those differences, researchers estimate how ancestral populations changed in size between about **10,000 and 1 million years ago**. [119] The result is an *effective population size*: a statistical measure of genetic variation, not a head count, and it depends on a model.

The diagram sketches what one study found. European and East Asian ancestors went through a severe bottleneck between about 10,000 and 60,000 years ago, around the time of the expansion out of Africa. African populations had a milder one and recovered earlier. All were larger between 60,000 and 250,000 years ago. [119]

That is a history of populations changing in size, not of a species appearing all at once. Because it measures genetic diversity rather than counting people, it cannot by itself show how few people lived at any one moment; page 43 turns to that question.

60–250 ka

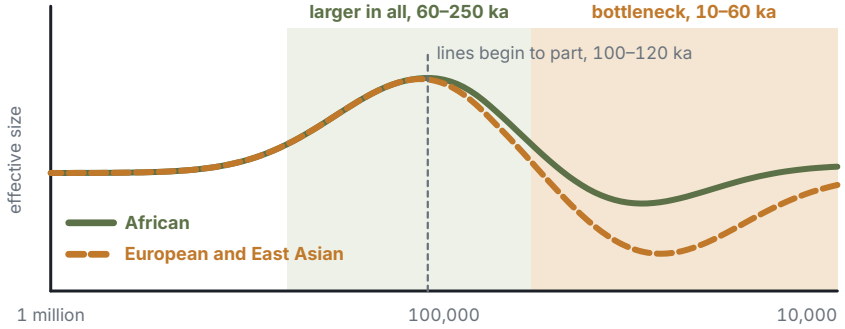
all ancestral populations larger

10–60 ka

a severe bottleneck in European and East Asian ancestors

1 genome

is enough to read this history



schematic: features from Li and Durbin 2011 [119], not their exact values

CHECK IT Li and Durbin 2011 **[119]**: peer-reviewed; no museum page

WAS THERE A FIRST HUMAN COUPLE?

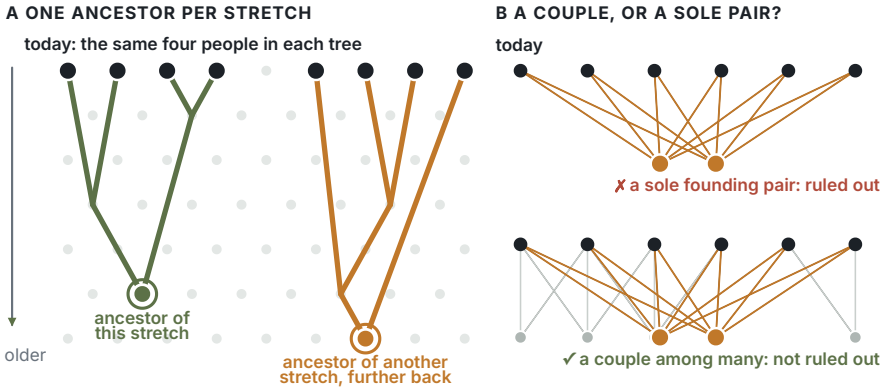
A common ancestor is not a first couple

Each stretch of our DNA traces back to a different ancestor. None of them was the first human, and none lived alone.

Traced back far enough, everyone's family trees overlap. Geneticists sometimes identify the *most recent common ancestor of one stretch of DNA*. That person was not the first human, nor alone: other stretches trace back to other people, at other times (diagram A).

"Mitochondrial Eve" and "Y-chromosomal Adam" are two such ancestors: of the DNA passed down only by mothers, and of the Y chromosome passed down only by fathers. One 2013 study places them about 99,000–148,000 and 120,000–156,000 years ago. [120] Each was one of thousands alive then; the labels do not describe a couple, and the two need never have met.

A recent sole founding pair? Could two people have been the *sole* biological founders of everyone alive today, with no other families and no later interbreeding? For roughly **the last 500,000 years**, our genes say no. [121, 122] A couple carries at most four versions of any gene, yet many human genes have far more versions than new mutations could have produced from one couple in that time. [121] That rules out a sole founding pair within that period, under ordinary inheritance. It does not rule out a couple among everyone's ancestors, alongside many others (diagram B). Geneticists accept this distinction. [121] It fits page 42: numbers rose and fell, with no sign of a narrowing to one pair.



grey dots: everyone else alive then

CHECK IT Poznik 2013 [120] · Swamidass 2019 [121, 122]

6

PART 6

The verdict

How strong is the case, and where is it uncertain?

Parts 4 and 5 ran the three tests set on page 14. This part gives the verdict: the results side by side, what is still debated, short answers to common questions and how to check any claim yourself. The box below says what the verdict means for scripture; Part 7 examines claims from revelation in detail.

WHAT THE VERDICT MEANS FOR SCRIPTURE

Judaism, Christianity and Islam share an account of origins: the world made in six days, and every human descended from one first couple, Adam and Eve (Genesis 1–2; Quran 7:54, 4:1). Read as history, a young Earth and a recent sole first couple are contradicted by the evidence in this booklet, all of it free for anyone to check: an Earth 4.54 billion years old (page 7), human relatives in dated rocks millions of years before our species (pages 29–34), and genes that did not pass through a sole founding pair in roughly the last 500,000 years (page 43). Believers who read these texts as theology rather than history can accept the science (page 51).

IN THIS PART

45 Did the three tests pass? · Different clues, one history
46 What is still uncertain? · Debate the branches, not the tree
47 What about…? · Short answers to common questions
48 How can I check it myself? · Follow one claim to its source

DID THE THREE TESTS PASS?

Different clues, one history

Fossils, dates and DNA use different methods and could have contradicted each other. They support the same broad history, with some details debated.

Page 14 set three tests. **What did we find?** All three held, by different methods on different materials, as the table below shows. Some tests concern shared ancestry itself; others, particular reconstructions such as where our species arose. **Could a separate creation explain the same details?** Only by making them look exactly as if humans had descended from earlier species: the same broken vitamin C gene and old virus DNA as other apes, in the same places, and a chromosome with a join where two ends fused (pages 38–39).

Among scientists, acceptance is overwhelming. In a 2014 survey by the Pew Research Center, 98% of members of the American Association for the Advancement of Science said humans evolved; among US adults, 65%. [123] Details are still debated (page 46), but the gap is between what the evidence has settled and what most people have heard.

COULD HAVE FOUND	ACTUALLY FOUND	
1 Test 1 · Older forms in older rocks		
✗ human skeletons in the oldest rocks	✓ **older forms in older layers**	p. 29
✗ big braincases from the start	✓ **braincases grow over time**	p. 33
✗ older *H. sapiens* outside Africa	✓ **oldest *H. sapiens* in Africa**	p. 34
2 Test 2 · A branching pattern		
✗ a DNA tree that contradicts anatomy	✓ **the same branching pattern**	p. 36
✗ Neanderthals no nearer than apes	✓ **Neanderthals far closer**	p. 40
3 Test 3 · Shared marks in our DNA		
✗ almost no shared broken genes	✓ **the same damage in place**	p. 38
✗ no sign of a join	✓ **telomere repeats at the join**	p. 39
Underneath all three: the dates		
✗ methods that disagree wildly	✓ **independent methods agree**	p. 26
✗ DNA and fossil dates at odds	✓ **broadly compatible dates**	p. 37

CHECK IT Summary of pages 26–43 · Pew 2015 [123]

WHAT IS STILL UNCERTAIN?

Debate the branches, not the tree

Scientists argue about the branches. They do not argue about whether there is a tree.

Researchers disagree about where some fossils fit, how ancestral African populations were connected, and exactly when groups moved and met. New dates, new bones and ancient DNA revise those details. This booklet has shown several open questions already: the Isua claim (page 8), Lucy's place (page 29), the Lomekwi tools (page 31) and the timing of Neanderthal mixing (page 40). [51, 64, 92, 110]

That is what a productive science looks like: a claim is exposed to tests that can correct it. It does **not** follow that humans might have appeared recently without evolutionary ancestors. To replace common ancestry, an alternative would need to explain the ordered fossil record, the dates, shared genomes, chromosome 2 and ancient DNA **at least as well**. The two columns below separate the trunk from the branches. [2, 27, 32, 33]

SETTLED
the trunk

- humans have evolutionary ancestors
- we share ancestors with other apes
- our species arose in Africa
- Neanderthal DNA is in living people
- chromosome 2 is a fusion
- apes share the same broken genes
- the Earth is 4.5 billion years old
- life is over 3.4 billion years old

STILL DEBATED
the branches

- where some fossils fit
- whether Lucy is our ancestor [51]
- how African populations were linked
- exact dates of dispersals
- when Australia was reached [115, 116]
- whether *Sahelanthropus* walked upright
- who made the Lomekwi tools [84]
- the oldest signs of life (Isua)

CHECK IT Bergström 2021 **[64]** · Ragsdale 2023 **[92]**

WHAT ABOUT…?

Short answers to common questions

"If humans evolved from apes, why are there still apes?"

Humans *are* apes. Our lineage and the chimpanzees' split from a shared ancestral population; one cousin does not disappear when another appears. See page 18. [27, 96]

"Where are the missing links?"

Fossilisation is rare, so the record has gaps, yet many forms with mixed traits are known: *Tiktaalik* (page 12), Lucy's species and *Homo erectus* (pages 29 and 32). [24, 28, 49]

"How do you know it was Africa?"

The oldest *H. sapiens* fossils and the greatest genetic diversity are both there. See pages 34 and 41. [34, 64, 75, 92]

"Isn't that just microevolution?"

Change within a population and the splitting of species are one process at different scales (page 11). [18, 23]

"Hasn't anyone actually seen it happen?"

Evolution, yes: the share of black peppered moths rose as the trees darkened and fell as they cleaned (page 10), and bacteria evolve resistance to antibiotics. New species form slowly, but the early stages, populations that stop interbreeding, have been observed forming (page 11). For the deep past, the record is in rocks and genomes. [17, 19, 23, 124]

"Isn't dating just assumptions?"

Different clocks, on different materials, based on different physics, give ages that agree within their stated errors. See pages 7 and 26. [6, 31]

"Doesn't shared DNA just show a common designer?"

A designer might reuse good parts. Shared *mistakes* are harder: humans and other apes carry the same broken vitamin C gene and old virus DNA in the same places (page 38). Inheritance predicts that; design does not. [96, 102]

"Is evolution the same as the origin of life?"

No: see page 9. *When* life began is dated on page 8; *how* is still open.

"But scripture says…"

Ask whether the text is being read as a physical claim (an age for the Earth, a first couple, a flood). If so, it can be tested, and Part 7 does exactly that (page 51). If not, the disagreement is not about evidence.

These are page references, not a script for winning an argument. Ask what observation would change the other person's mind, and invite them to check the source.

HOW CAN I CHECK IT MYSELF?

Follow one claim to its source

You need not accept a diagram on authority. Pick any claim in this booklet and follow it to its source, in the four steps below.

Take Lucy, card 3 on page 27. The four cards below follow her from a catalogue number to a test.

"But AI or a website could have made this up." Then check it the same way. Almost every paper on pages 62–66 has a DOI you can type into any browser; books are named in full, and museum pages have their web address.

Details will change with new evidence. The central conclusion rests on several independent kinds of evidence that agree. Checking a claim before believing it is ordinary due diligence. The next part asks what happens when the same test is applied to claims that rest on authority instead.

1 **Find** a specimen number	Lucy's catalogue number is AL 288-1. Search it in the Smithsonian's online fossil catalogue: it gives the find site, Hadar in Ethiopia, and the species. [49]
2 **Date** the layer, with its error	Ask how the age was measured and how large its uncertainty is. Lucy is about 3.2 million years old; page 26 shows how an age is cross-checked, at a different site.
3 **Compare** with other evidence	Open the Natural History Museum's page on Lucy's species. Does it say the same? Page 60 lists the museum page behind each key claim. [73]
4 **Test** what each idea predicts	If Lucy's species shares ancestry with us, its bones should mix older and newer traits. They do: a pelvis for walking upright, long arms for climbing (page 29).

CHECK IT All sources: pages 60–66

7

PART 7

Evidence and authority

How do other ways of knowing compare?

The evidence is in. This last part applies the same method to claims that rest on authority. Literal claims about nature can be tested like any other, and the five tested here fail; for the rest, no evidence that stands up to scientific testing has been presented. Test the argument the same way. It ends with the author's own view.

IN THIS PART

50 How should a belief be held? · A claim that can be corrected, and one that cannot
51 Can scripture's physical claims be tested? · Five claims, tested
52 Is revelation a way of knowing? · Certainty is not the same as being right
53 What are the best objections? · Objections, answered
54 Afterword · Why I wrote this

HOW SHOULD A BELIEF BE HELD?

A claim that can be corrected, and one that cannot

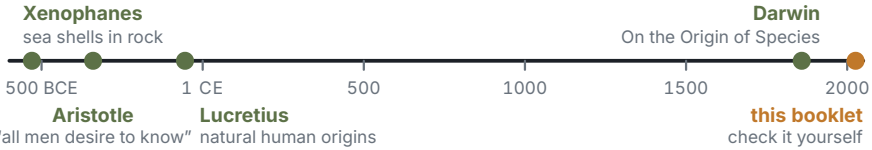
A belief that nothing could ever show to be wrong cannot tell us what happened.

Everything on the evidence pages can be checked, and any of it could be shown wrong; some details already have been (page 46). This is the scientific method, and it is simply due diligence: ask how a claim could be checked, look at the evidence, hold the belief in proportion to it, and change it when the evidence changes. It works as well for a fossil as for a medicine or a claim in the news.

Faith is the opposite. Here, faith means holding a claim about the physical world because an authority states it, and letting no observation count against it. That has no way of reaching an answer, because nothing can correct it. Trusting an expert is different: a museum's claim comes with a catalogue number and a method, and would change if the evidence did.

What the method says about religion. Religious claims are treated like any other claim. Literal statements about the physical world (a young Earth, a first couple, a flood over the highest mountains) can be tested, and the five on page 51 fail. For a god itself, no evidence that stands up to scientific testing has been presented, so science has nothing to support the claim, as with any other claim made without evidence. If such evidence appeared, the method would require it to be examined and followed. Many believers read scripture as poetry or theology and accept evolution; none of this is aimed at them.

An old habit. Asking for evidence is not new; the timeline below spans 2,500 years. Xenophanes saw sea shells in mountain rock in the sixth century BCE and concluded there had once been a sea there. Aristotle began his *Metaphysics* with "All men by nature desire to know". Lucretius gave a natural account of human origins in the first century BCE. [125] They share the habit this booklet asks of you: look at the evidence, not only at who is speaking.



CAN SCRIPTURE'S PHYSICAL CLAIMS BE TESTED?

Five claims, tested

Read literally, some passages of sacred texts make claims about the physical world, and those claims can be tested like any other.

The table below gives five, each in its literal form; the same test applies to any other.

Reading non-literally. Many believers read these passages as poetry or theology and accept evolution. [3] Augustine read the six days of Genesis non-literally around AD 400 [126]; other such readings came after science settled a question. Such a reading makes no physical claim to test.

TEXT	READ LITERALLY	WHAT THE EVIDENCE SHOWS
Genesis 1; Ussher's date	The world was made a few thousand years ago; Ussher dated it to 4004 BCE [127]	The Earth is 4.54 billion years old, about 750,000 times older [6]
Genesis 2; Quran 4:1; Romans 5	All humans descend from one created couple, the only people of their time	No sign of a sole founding pair in roughly the last 500,000 years (page 43) [121]
Genesis 6–9	A flood covered the highest mountains, killing every land animal outside the ark	Antarctic ice cores give a continuous, dated record of 800,000 years, which a recent worldwide flood would have disrupted [128]
Joshua 10:12–13; Psalm 93:1	The Earth stands still and the Sun moves round it; at Joshua's command the Sun stopped for about a day	The Earth spins daily and orbits the Sun yearly: Foucault's pendulum (1851) shows the spin [129]; the star shift Bessel measured in 1838 shows the orbit [130]
Book of Mormon, 1981 introduction	Israelites who sailed to the Americas about 600 BCE were the principal ancestors of the American Indians	Most Indigenous American ancestry traces to Siberia, over 15,000 years ago [118, 131]. In 2006 the Church changed "principal" to "among the" [132]

CHECK IT Ussher 1650 [127] · Swamidass 2019 [121] · Skoglund and Reich 2016 [118] · Southerton 2004 [131] · Deseret News 2007 [132] · Jouzel 2007 [128] · Sommeria 2017 [129] · ESA [130]

IS REVELATION A WAY OF KNOWING?

Certainty is not the same as being right

Feeling certain is not evidence of being right.

The deeper question is whether revelation, meaning knowledge said to come from a sacred text, a prophet or an inner conviction, can on its own tell us what happened in the world. Five reasons suggest it cannot.

1. Revelations contradict each other. The Gospels say Jesus died on the cross. The Quran says he was not killed or crucified (4:157). Hindu cosmology describes endless cycles of creation; Genesis describes a single beginning. Believers in each hold their view with the same deep certainty. That does not show every revelation is false; it shows that certainty alone cannot decide between them.

2. It offers no reliable method of prediction. Science states in advance what should be found, and risks being wrong: *Tiktaalik* (page 12) was predicted before it was found, and the join in chromosome 2 (page 39) could have been absent. [24, 33] Claims that scripture foretold modern science are usually made *after* the discovery, by re-reading old words to fit it.

3. Where its physical claims conflict with science, they fail. Sacred texts contain much ordinary history and geography that is true. But where a literal claim about nature contradicts established science, as in the examples on page 51, the evidence has gone against it.

4. A claim that nothing could disprove cannot guide inquiry. "God guided evolution in ways we cannot detect" fits every possible observation. It may carry meaning for a believer, but it predicts nothing that would tell one physical history from another, so science cannot use it.

5. How strongly we feel is not evidence. The day after the 11 September 2001 attacks, US students wrote down how they heard the news. Months later many accounts had changed, yet the students felt just as confident in them. [133] Confidence tracks how strongly we believe, not whether what we believe is true.

> *"The first principle is that you must not fool yourself, and you are the easiest person to fool."*
>
> Richard Feynman, 1974 [134]

CHECK IT Daeschler, Shubin and Jenkins 2006 [24] · Talarico and Rubin 2003 [133] · Feynman 1974 [134]

WHAT ARE THE BEST OBJECTIONS?

Objections, answered

An argument is only as good as its answer to the best case against it.

"Science is just another faith."

Everyone relies on trust, scientists included. The difference is what it rests on: scientific claims can be checked by anyone with the means, and change when the evidence does (page 46). Science catches its own frauds: Piltdown Man (1912) was exposed in 1953 when fluorine tests showed the bones were far too young. [135] Testing can overturn a claim; a belief nothing can overturn has no check.

"You can't prove there is no god."

True, and science does not try: nothing can prove the absence of everything imaginable, such as invisible dragons or a teapot orbiting the Sun. The claimant carries the burden of proof; until evidence arrives, belief waits.

"Science can't explain everything."

True: open questions are listed on page 46. But a gap in knowledge is not evidence for any particular answer. Lightning, disease and the origin of species were once explained by gods; each proved to have a natural explanation, and no supernatural one ever has.

"God made the world look old."

Then light from Supernova 1987A, 168,000 light-years away, shows a star that never existed exploding. [136] So would every rock layer, fossil and gene. A claim that fits every possible observation cannot be tested (page 52), and would make the Creator the author of a false record.

"Many great scientists were religious."

True: Gregor Mendel was a friar and Georges Lemaître (the "primeval atom") a priest; Francis Collins is an evangelical Christian. Their discoveries stand because they could be checked, not through faith.

"Without God there is no basis for morality."

Two claims hide here. That people need religion to be good is answered by experience: honesty, fairness and care are found in every culture, among believers and non-believers alike. That moral duties need a divine source is a serious philosophical argument. The afterword sets out one view.

In short. When we want to know what is true about the world, evidence has earned our trust because it can be checked and corrected; revelation has not. That says nothing against the worth of religious communities, art or comfort.

AFTERWORD

Why I wrote this

The rest of this booklet reports what the evidence shows. This page is different: it is my own view, kept apart so that the rest can be quoted without it.

I do not think religion is a reliable way of finding out what is true. Its claims about the physical world were made with complete confidence: a young Earth, a first couple, a flood over the highest mountains. Where those claims can be checked, they have failed (page 51). Faith is not a better method that happened to lose these cases. It is the absence of one: nothing a believer could observe is allowed to count against the belief.

Revelation has the same flaw. Scriptures and prophets contradict one another, each of them certain, and none offers a way to tell which, if any, is right (page 52). A source that cannot be checked or corrected is not knowledge, however old it is and however sincerely it is held.

Nor do we need it to be good. A book offered as the foundation of morality also sets out rules for buying slaves and keeping them for life (Leviticus 25:44–46). Readers who reject those verses are judging scripture by a moral sense they already have. I am a humanist, in the sense of the Amsterdam Declaration of 2022: morality grows out of the capacity of living things to suffer and to flourish, and needs no source outside humanity. [137]

None of this is a judgement on believers, many of them kinder and wiser than I am. It is a judgement on a way of knowing, and it applies to me too: if I exempted my own view from testing, I could not ask anyone to test theirs. So this booklet asks you to believe nothing on my authority. It gives you specimen numbers, dates with error bars and the identifiers of the papers, and asks you to look.

This booklet is also an experiment in using AI well. The question, the argument and every decision are mine; AI helped me research, draft, draw and check, far faster than I could alone (page 58). It can be wrong, which is why every claim carries a source you can check.

What I hope you take from it is a habit: when someone tells you what happened in the world, or how you must live, ask how they know, what would change their mind, and where you can check.

George Philippou
September 2026

REFERENCE

Check it yourself

Where can you check every claim?

Everything in this booklet can be checked. Here are the tools: a glossary, further reading, a table matching key claims to museum pages, every source with its DOI or web address, and an index.

IN THIS PART

56–57 Glossary · Words used in this booklet
58 Further reading · If you want to go further
59 Image credits · Where the pictures come from
60 The cross-check · Checked against the museums
62–66 Check the evidence · Sources you can inspect
67 Index · Specimens, sites and key terms

GLOSSARY

Words used in this booklet

EVOLUTION AND CLASSIFICATION

Abiogenesis The origin of life from non-living chemistry. A separate question from evolution, which explains how life changed once it existed; still an open field of research (page 9).

Archaea, bacteria, eukaryotes The three great branches of the tree of life. Bacteria and archaea are single cells without a nucleus; eukaryotes, which include plants, fungi and animals, have one, and grew out of the archaea (page 15).

Bipedal Walking upright on two legs (pages 29–30).

Denisovans An extinct human group first identified in 2010 from the DNA of a finger bone. Few fossils are known: teeth, a jaw from Tibet, and the Harbin skull from China, identified as Denisovan in 2025 (pages 19, 20).

Effective population size A model-based measure of genetic variation. Not a headcount (page 42).

Genetic drift Random change in how common a gene variant is, from one generation to the next, by luck rather than selection; strongest in small populations (page 11).

Genus A group of closely related species. *Homo* is our genus; *Homo sapiens* our species.

Hominid, hominin Hominids are all great apes, living and extinct, humans included. Hominins are only the human branch after it split from the line leading to chimpanzees and bonobos (page 18).

Natural selection Traits that improve survival or reproduction in an environment become more common over generations, because their carriers leave more descendants (page 10).

Reproductive isolation A barrier that stops two populations interbreeding; its gradual evolution is how one population becomes two species (page 11).

Species In practice, a population whose members can interbreed and produce fertile offspring. Boundaries blur while populations are diverging, so “the moment a species appeared” is rarely well defined.

DATING AND DEEP TIME

Cosmic microwave background Faint radiation from the young universe, released about 380,000 years after the start (page 6).

Electron spin resonance Measures radiation damage built up in tooth enamel to give its age.

Ma, ka Million years ago; thousand years ago.

Radioactive dating Any method based on the steady decay of unstable atoms into stable ones, such as uranium into lead (page 7). Carbon-14 is one, useful only to about 50,000 years.

Rock layers (stratigraphy) In undisturbed deposits, a lower layer is older than the one above it (page 26).

Stromatolite A layered mound built by microbial mats; the oldest secure evidence of life (page 8).

Thermoluminescence Dates once-heated minerals, such as burnt flint, from the light they release when reheated (page 26).

Zircon A tough mineral that locks in uranium as it forms; the oldest known pieces of the Earth.

GLOSSARY

Words used in this booklet, continued

GENETICS AND INHERITANCE

Ancient DNA Genetic material recovered from old remains, such as Neanderthal bones, and sequenced despite damage and contamination.

Broken gene (pseudogene) A gene that no longer works, such as our vitamin C gene (page 38).

Centromere, telomere The pinched waist of a chromosome and its protective ends. Their distinctive sequences are why "ends" in the middle of chromosome 2 are a clue (page 39).

Chromosome One long DNA molecule with its packaging. Humans have 23 pairs.

Genome An organism's complete set of DNA.

Molecular clock Dating a split from the number of DNA differences between two species and the rate at which differences arise (page 37).

Most recent common ancestor For one stretch of DNA, the ancestor in whom all present copies meet. Different stretches have different ancestors (page 43).

Mutation A change in a DNA sequence, arising during copying or repair. Most have no effect; mutations are the raw material that inheritance and selection act on.

Old virus DNA (endogenous retrovirus) The remains of a virus that copied itself into the DNA of an egg or sperm cell and was then inherited (page 38).

STONE TOOLS

Oldowan, Acheulean The earliest widespread toolkits: sharp flakes struck from cores, from about 2.6 million years ago, then shaped handaxes, from about 1.76 million (page 31).

HOW SCIENCE WORKS

Fact, theory A fact is an observation or a finding beyond reasonable doubt; a theory is the tested explanation that ties facts together and predicts new ones (page 5).

Hypothesis A specific, testable explanation, precise enough that a prediction drawn from it could turn out wrong (page 5).

Peer review Independent checking of a study by other specialists before publication. It filters errors; it does not guarantee a result, which is why independent replication matters.

Scientific consensus The position most scientists in a field reach after weighing the accumulated evidence: not a vote, but a convergence of independent lines of evidence (page 45).

Scientific law A concise description of a pattern, such as Newton's law of gravitation. A law describes what happens; a theory explains why (page 5).

Testable (falsifiable) A claim is testable only if some possible observation would count against it. A claim that fits every outcome cannot be tested (page 52).

FURTHER READING

If you want to go further

Everything in this booklet can be checked on pages 60–66. These are the places to go for the wider story.

VISIT

Natural History Museum, London, Human Evolution gallery: Gibraltar 1 and Cheddar Man are on display, with casts of many specimens in this booklet. **Smithsonian National Museum of Natural History**, Washington, DC: the David H. Koch Hall of Human Origins.

HUMAN ORIGINS

Louise Humphrey and Chris Stringer, *Our Human Story* (Natural History Museum, 2018): short and illustrated, by two of the Museum's researchers. Svante Pääbo, *Neanderthal Man* (2014): the first-hand story of ancient DNA. Adam Rutherford, *A Brief History of Everyone Who Ever Lived* (2016).

EVOLUTION IN GENERAL

Jerry Coyne, *Why Evolution Is True* (2009). Neil Shubin, *Your Inner Fish* (2008): the search for *Tiktaalik*. Richard Dawkins, *The Ancestor's Tale* (2004; revised 2016).

THE ORIGIN OF LIFE

A separate question from evolution (page 9). Nick Lane, *The Vital Question* (2015): energy, cells and how life may have begun. Addy Pross and Robert Pascal, "The origin of life: what we know, what we can know and what we will never know", *Open Biology* (2013), free online. [14] Online: the Natural History Museum's *The origin of life on Earth* [9] and the "Origin of Life" module at evolution.berkeley.edu.

DEEP TIME

G. Brent Dalrymple, *The Age of the Earth* (1991). Simon Singh, *Big Bang* (2004).

CLASSICAL SOURCES

Lucretius, *The Nature of Things*, tr. A. E. Stallings (Penguin, 2007). Aristotle, *Metaphysics*, tr. W. D. Ross (1924; free online).

ONLINE

nhm.ac.uk/discover: the Museum's human-evolution articles. **humanorigins.si.edu**: a searchable fossil catalogue with 3-D scans. **evolution.berkeley.edu**: the concepts explained. **europepmc.org**: free abstracts of the papers cited here.

HOW THIS BOOKLET WAS MADE

Conceived and directed by George Philippou, who made every final decision and reviewed every page. AI (Claude, by Anthropic) was the tool for research, drafting, diagrams, layout and checking claims against sources. The sources are the authority: check them.

IMAGE CREDITS

Where the pictures come from

Every photograph, drawing and map in this booklet, with its author and licence.

All diagrams and maps are original, © the author. The illustrations on pp. 12 (the three animals), 26 (dating diagram), 29 (leg diagram) and 31 (stone tools) are © the author, created with AI image tools; text, drawings and layout were prepared with AI assistance. Photographs, via Wikimedia Commons, are credited in their captions; in addition, specimen photos (p. 27): Toumaï cast, Didier Descouens, CC BY-SA 4.0; *Ardipithecus* skull cast, Chris Woodrich, CC BY-SA 4.0; Lucy, "120", CC BY 2.5; KNM-ER 3733 replica, Gerbil, CC BY-SA 3.0; La Chapelle-aux-Saints 1, Luna04, CC BY 2.5. Family tree photographs (p. 19): Jebel Irhoud 1 cast, Jonathan Chen, CC BY-SA 4.0; La Chapelle-aux-Saints 1, Luna04, CC BY 2.5; KNM-ER 3733 replica, Gerbil, CC BY-SA 3.0; KNM-ER 1813, José-Manuel Benito Álvarez, public domain; *Paranthropus boisei* cast, Daderot, CC0; Lucy, "120", CC BY 2.5; *Ardipithecus* skull cast, Chris Woodrich, CC BY-SA 4.0; Harbin cranium, as for the *Homo* chart. *Homo* chart (pp. 20–21): SK 48, *Paranthropus robustus*, José Braga and Didier Descouens, CC BY-SA 4.0; Kabwe 1 (Broken Hill skull cast), Stanislav Kozlovskiy, CC BY-SA 4.0; Harbin cranium (Denisovan; also p. 19), Fu et al. 2025, *Cell*, Fig. 1A, CC BY 4.0, background removed; other skulls as credited on pp. 22–23 and 27. Coastlines from Natural Earth (public domain). Fossil ranges from the Smithsonian Human Origins Program. Specimen facts from the Smithsonian and NHM pages cited.

THE CROSS-CHECK (1/2)

Checked against the museums

Each key claim, and the museum page that says the same. Sources [n]: pages 62–66.

P.	CLAIM	NATURAL HISTORY MUSEUM	SMITHSONIAN
7–8	Earth ~4.5 billion years old; life by ~3.5 billion	Origin of life [9]; Oldest life [13]	—
11	Isolation puts populations on the path to new species	What is a species? [22]	—
17–18	Humans and chimps share an ancestor, 9–6 million years ago (a genetic estimate)	Modern humans [35]	Genetics [47]
19–20	A tree, not a chain; several *Homo* species lived at once	Lucy's species [73]; 2025 news [51]; *Homo naledi* [52]; Denisovans [54]	Species [28]; Family tree [29]
26, 34	*H. sapiens* in Africa by ~300,000 years ago, dated two ways	Jebel Irhoud [68]; Modern humans [35]	Dating [67]; *Homo sapiens* [95]
27–28	Catalogued specimens; Gibraltar 1; Dmanisi	First adult Neanderthal skull [69]; *Homo erectus* [77]	Fossil catalogue [49]
29–30	*A. afarensis* walked upright; Laetoli footprints, 3.6–3.7 Ma	Lucy's species [73]	Species [28]; Laetoli [30]
31–33	Tools by 2.6 Ma; *H. erectus*; braincase sizes	*Homo erectus* [77]; *Homo naledi* [52]	Stone tools [55]; Species [28]; Fossil catalogue [49]
36	"% similar" depends on what is counted	—	Genetics [47]
40–41	Neanderthal and Denisovan DNA; mixing 45–49 ka; Cheddar Man	Neanderthals [109]; Denisovans [54]; 2024 news [111]; Cheddar Man [114]	Ancient DNA [112]

THE CROSS-CHECK (2/2)

Checked against papers and academies

No museum page covers these. They rest on peer-reviewed research or the US National Academies. Sources [n]: pages 62–66.

P.	CLAIM	SOURCE
4–6	Satellite clocks; theory and law; age of the universe	Ashby 2003 [1]; UC Berkeley [4]; US National Academies [2, 3]; Planck 2020 [5]
10–12	The peppered moth; selection in humans; the *Tiktaalik* prediction	van 't Hof 2016 [16]; Cook and Saccheri 2013 [17]; Tishkoff 2007 [65]; Daeschler 2006 [24]
15–16	The tree of life; nested groups from animals to apes	Hug 2016 [36]; Eme 2023 [38]; Bobrovskiy 2018 [39]; Shu 1999 [40]; Ni 2013 [42]; Stevens 2013 [43]
24	What is distinctive about humans: culture, language, cooperation, social cognition, life history, hands	Henrich 2016 [58]; Hauser 2002 [59]; Fehr 2003 [60]; Herrmann 2007 [61]; Hawkes 1998 [62]; Marzke 2013 [63]; Laland 2010 [66]
37	DNA clock dates the human–chimp split	Langergraber 2012 [98]; Moorjani 2016 [99]; Yoo 2025 [97]
38	Shared broken gene; shared virus insertions	Ohta and Nishikimi 1999 [102]; Mansueto 2024 [103]; Chimpanzee Consortium 2005 [96]
39, 42–43	Chromosome 2 fusion; population sizes; no sole founding pair for ~500,000 years	IJdo 1991 [33]; NHGRI [106]; Li and Durbin 2011 [119]; Swamidass 2019 [121, 122]
45	What scientists and the public think	Pew Research Center 2015 [123]

Pages 50–54 combine sourced factual claims with philosophical argument and personal opinion; no museum or academy endorses the argument. Museum pages were opened and compared on 23 Sep 2026; web pages can change.

CHECK THE EVIDENCE (1/5)

Sources you can inspect

Type any DOI after **https://doi.org/** in a browser to reach the paper. Museum pages are numbered in colour: Natural History Museum in orange (add **https://www.** in front of the address), Smithsonian in blue (add **https://**).

[1] Ashby, N. (2003) Relativity in the Global Positioning System. *Living Reviews in Relativity* 6:1. doi:10.12942/lrr-2003-1

[2] US National Academies, *Evolution Resources: Is evolution a theory or a fact?* nationalacademies.org/evolution-resources

[3] National Academy of Sciences and Institute of Medicine (2008) *Science, Evolution, and Creationism*. National Academies Press. doi:10.17226/11876

[4] UC Museum of Paleontology, *Understanding Science* glossary: *theory* and *law*. undsci.berkeley.edu/glossary/theory/ and undsci.berkeley.edu/glossary/law/

[5] Planck Collaboration (2020) Planck 2018 results. VI. Cosmological parameters. *Astronomy & Astrophysics* 641:A6. doi:10.1051/0004-6361/201833910

[6] Dalrymple (2001) The age of the Earth in the twentieth century: a problem (mostly) solved. *Geol. Soc. London Spec. Publ.* 190:205–221. doi:10.1144/GSL.SP.2001.190.01.14

[7] Wilde et al. (2001) Evidence from detrital zircons for continental crust and oceans on the Earth 4.4 Gyr ago. *Nature* 409:175–178. doi:10.1038/35051550

[8] National Academies of Sciences, Engineering, and Medicine (2016) *Molybdenum-99 for Medical Imaging*. National Academies Press. doi:10.17226/23563

[9] NHM, *The origin of life on Earth*. nhm.ac.uk/discover/origin-of-life-on-earth.html

[10] Allwood et al. (2006) Stromatolite reef from the Early Archaean era of Australia. *Nature* 441:714–718. doi:10.1038/nature04764

[11] Nutman et al. (2016) Rapid emergence of life shown by discovery of 3,700-million-year-old microbial structures. *Nature* 537:535–538. doi:10.1038/nature19355

[12] Allwood et al. (2018) Reassessing evidence of life in 3,700-million-year-old rocks of Greenland. *Nature* 563:241–244. doi:10.1038/s41586-018-0610-4

[13] NHM news (2022) *The oldest evidence of life on Earth could help discover ancient life on Mars*. nhm.ac.uk/discover/news/2022/november/oldest-evidence-life-earth-could-help-discover-ancient-life-mars.html

[14] Pross and Pascal (2013) The origin of life: what we know, what we can know and what we will never know. *Open Biology* 3:120190. doi:10.1098/rsob.120190

[15] Kong et al. (2012) Rate of de novo mutations and the importance of father's age to disease risk. *Nature* 488:471–475. doi:10.1038/nature11396

[16] van 't Hof et al. (2016) The industrial melanism mutation in British peppered moths is a transposable element. *Nature* 534:102–105. doi:10.1038/nature17951

[17] Cook and Saccheri (2013) The peppered moth and industrial melanism: evolution of a natural selection case study. *Heredity* 110:207–212. doi:10.1038/hdy.2012.92

[18] UC Museum of Paleontology, *Defining microevolution* and *What is macroevolution?* evolution.berkeley.edu/evolution-101/microevolution/defining-microevolution/ and evolution.berkeley.edu/evolution-101/macroevolution/what-is-macroevolution/

[19] Cook, Dennis and Mani (1999) Melanic morph frequency in the peppered moth in the Manchester area. *Proc. R. Soc. B* 266:293–297. doi:10.1098/rspb.1999.0636

[20] Cook et al. (2012) Selective bird predation on the peppered moth: the last experiment of Michael Majerus. *Biology Letters* 8:609–612. doi:10.1098/rsbl.2011.1136

[21] UC Museum of Paleontology, *Understanding Evolution*: *Genetic drift*. evolution.berkeley.edu/evolution-101/mechanisms-the-processes-of-evolution/genetic-drift/

[22] NHM, *What is a species, and how many species are there?* nhm.ac.uk/discover/what-is-a-species.html

[23] Hsu et al. (2024) Reproductive isolation arises during laboratory adaptation to a novel hot environment. *Genome Biology* 25:141. doi:10.1186/s13059-024-03285-9

[24] Daeschler, Shubin and Jenkins (2006) A Devonian tetrapod-like fish and the evolution of the tetrapod body plan. *Nature* 440:757–763. doi:10.1038/nature04639

[25] Shubin, N. (2008) *Your Inner Fish: A Journey into the 3.5-Billion-Year History of the Human Body*. Pantheon.

[26] Niedźwiedzki et al. (2010) Tetrapod trackways from the early Middle Devonian period of Poland. *Nature* 463:43–48. doi:10.1038/nature08623

[27] Smithsonian Institution, *Human Evolution Evidence*. humanorigins.si.edu/evidence

[28] Smithsonian Institution, *Early human species* (species pages and time ranges). humanorigins.si.edu/evidence/human-fossils/species

CHECK THE EVIDENCE (2/5)

Sources you can inspect

[29] Smithsonian Institution, *Human Family Tree*. humanorigins.si.edu/evidence/human-family-tree

[30] Smithsonian Institution, *Laetoli Footprint Trails*. humanorigins.si.edu/evidence/behavior/footprints

[31] Richter et al. (2017) The age of the hominin fossils from Jebel Irhoud, Morocco. *Nature* 546:293–296. doi:10.1038/nature22335

[32] Green et al. (2010) A draft sequence of the Neandertal genome. *Science* 328:710–722. doi:10.1126/science.1188021

[33] IJdo et al. (1991) Origin of human chromosome 2: an ancestral telomere–telomere fusion. *PNAS* 88:9051–9055. doi:10.1073/pnas.88.20.9051

[34] Hublin et al. (2017) New fossils from Jebel Irhoud, Morocco and the pan-African origin of *Homo sapiens*. *Nature* 546:289–292. doi:10.1038/nature22336

[35] NHM, *Modern humans, Homo sapiens: when, where and how did we evolve?* nhm.ac.uk/discover/modern-humans-homo-sapiens-when-where-how-did-we-evolve.html

[36] Hug et al. (2016) A new view of the tree of life. *Nature Microbiology* 1:16048. doi:10.1038/nmicrobiol.2016.48

[37] Woese and Fox (1977) Phylogenetic structure of the prokaryotic domain: the primary kingdoms. *PNAS* 74:5088–5090. doi:10.1073/pnas.74.11.5088

[38] Eme et al. (2023) Inference and reconstruction of the heimdallarchaeial ancestry of eukaryotes. *Nature* 618:992–999. doi:10.1038/s41586-023-06186-2

[39] Bobrovskiy et al. (2018) Ancient steroids establish the Ediacaran fossil *Dickinsonia* as one of the earliest animals. *Science* 361:1246–1249. doi:10.1126/science.aat7228

[40] Shu et al. (1999) Lower Cambrian vertebrates from south China. *Nature* 402:42–46. doi:10.1038/46965

[41] NHM, *Twelve treasures of our gardens* (*Megazostrodon*, one of the earliest known mammals, 200 million years ago). nhm.ac.uk/discover/garden-highlights.html

[42] Ni et al. (2013) The oldest known primate skeleton and early haplorhine evolution. *Nature* 498:60–64. doi:10.1038/nature12200

[43] Stevens et al. (2013) Palaeontological evidence for an Oligocene divergence between Old World monkeys and apes. *Nature* 497:611–614. doi:10.1038/nature12161

[44] UC Museum of Paleontology, *Understanding Evolution*: *Homologies and analogies*. evolution.berkeley.edu/evolution-101/the-history-of-life-looking-at-the-patterns/homologies-and-analogies/

[45] Xia et al. (2024) On the genetic basis of tail-loss evolution in humans and apes. *Nature* 626:1042–1048. doi:10.1038/s41586-024-07095-8

[46] Tojima et al. (2018) Tail reduction process during human embryonic development. *Journal of Anatomy* 232:806–811. doi:10.1111/joa.12774

[47] Smithsonian Institution, *Genetics*. humanorigins.si.edu/evidence/genetics

[48] Darwin, C. (1871) *The Descent of Man, and Selection in Relation to Sex*, chapter 6. John Murray, London. Free online: darwin-online.org.uk

[49] Smithsonian Institution, *Human Fossils* catalogue (search the IDs on p. 27). humanorigins.si.edu/evidence/human-fossils/fossils

[50] Fu et al. (2025) Denisovan mitochondrial DNA from dental calculus of the >146,000-year-old Harbin cranium. *Cell* 188:3919–3926. doi:10.1016/j.cell.2025.05.040

[51] NHM news (2025) *Early hominin Australopithecus afarensis may not be our human ancestor after all*. nhm.ac.uk/discover/news/2025/november/early-hominin-australopithecus-afarensis-may-not-be-our-human-ancestor.html

[52] NHM, *Homo naledi, your recently discovered human relative*. nhm.ac.uk/discover/homo-naledi-your-most-recently-discovered-human-relative.html

[53] Reich et al. (2010) Genetic history of an archaic hominin group from Denisova Cave. *Nature* 468:1053–1060. doi:10.1038/nature09710

[54] NHM, *Denisovans: the enigmatic ancient humans who finally have a face*. nhm.ac.uk/discover/denisovans.html

[55] Smithsonian Institution, *Stone Tools* and *Early Stone Age Tools*. humanorigins.si.edu/evidence/behavior/stone-tools

[56] Berger et al. (2015) *Homo naledi*, a new species of the genus *Homo* from the Dinaledi Chamber, South Africa. *eLife* 4:e09560. doi:10.7554/eLife.09560

[57] Sutikna et al. (2016) Revised stratigraphy and chronology for *Homo floresiensis* at Liang Bua in Indonesia. *Nature* 532:366–369. doi:10.1038/nature17179

CHECK THE EVIDENCE (3/5)

Sources you can inspect

[58] Henrich, J. (2016) *The Secret of Our Success: How Culture Is Driving Human Evolution, Domesticating Our Species, and Making Us Smarter*. Princeton University Press. Cumulative cultural evolution and the "ratchet effect": pp. 54–57.

[59] Hauser, Chomsky and Fitch (2002) The faculty of language: what is it, who has it, and how did it evolve? *Science* 298:1569–1579. doi:10.1126/science.298.5598.1569

[60] Fehr and Fischbacher (2003) The nature of human altruism. *Nature* 425:785–791. doi:10.1038/nature02043

[61] Herrmann et al. (2007) Humans have evolved specialized skills of social cognition: the cultural intelligence hypothesis. *Science* 317:1360–1366. doi:10.1126/science.1146282

[62] Hawkes et al. (1998) Grandmothering, menopause, and the evolution of human life histories. *PNAS* 95:1336–1339. doi:10.1073/pnas.95.3.1336

[63] Marzke, M. W. (2013) Tool making, hand morphology and fossil hominins. *Phil. Trans. R. Soc. B* 368:20120414. doi:10.1098/rstb.2012.0414

[64] Bergström et al. (2021) Origins of modern human ancestry. *Nature* 590:229–237. doi:10.1038/s41586-021-03244-5

[65] Tishkoff et al. (2007) Convergent adaptation of human lactase persistence in Africa and Europe. *Nature Genetics* 39:31–40. doi:10.1038/ng1946

[66] Laland, Odling-Smee and Myles (2010) How culture shaped the human genome: bringing genetics and the human sciences together. *Nature Reviews Genetics* 11:137–148. doi:10.1038/nrg2734

[67] Smithsonian Institution, *Dating*. humanorigins.si.edu/evidence/dating

[68] NHM news (2017) *Oldest known Homo sapiens fossils discovered in Morocco*. nhm.ac.uk/discover/news/2017/june/oldest-known-homo-sapiens-fossils-discovered-in-morocco.html

[69] NHM, *First adult Neanderthal skull* (Gibraltar 1). nhm.ac.uk/discover/first-adult-neanderthal-skull.html

[70] Lebatard et al. (2008) Cosmogenic nuclide dating of *Sahelanthropus tchadensis*. *PNAS* 105:3226–3231. doi:10.1073/pnas.0708015105

[71] Daver et al. (2022) Postcranial evidence of late Miocene hominin bipedalism in Chad. *Nature* 609:94–100. doi:10.1038/s41586-022-04901-z; and Macchiarelli et al. (2020) Nature and relationships of *Sahelanthropus tchadensis*. *J. Human Evolution* 149:102898. doi:10.1016/j.jhevol.2020.102898

[72] Williams et al. (2026) Earliest evidence of hominin bipedalism in *Sahelanthropus tchadensis*. *Science Advances* 12:eadv0130. doi:10.1126/sciadv.adv0130

[73] NHM, *Australopithecus afarensis, Lucy's species*. nhm.ac.uk/discover/australopithecus-afarensis-lucy-species.html

[74] Lepre and Kent (2015) Chronostratigraphy of KNM-ER 3733 and other Area 104 hominins from Koobi Fora. *J. Human Evolution* 86:99–111. doi:10.1016/j.jhevol.2015.06.010

[75] Vidal et al. (2022) Age of the oldest known *Homo sapiens* from eastern Africa. *Nature* 601:579–583. doi:10.1038/s41586-021-04275-8

[76] Ferring et al. (2011) Earliest human occupations at Dmanisi dated to 1.85–1.78 Ma. *PNAS* 108:10432–10436. doi:10.1073/pnas.1106638108

[77] NHM, *Homo erectus, our ancient ancestor*. nhm.ac.uk/discover/homo-erectus-our-ancient-ancestor.html

[78] Ward, Kimbel and Johanson (2011) Complete fourth metatarsal and arches in the foot of *Australopithecus afarensis*. *Science* 331:750–753. doi:10.1126/science.1201463

[79] Raichlen et al. (2010) Laetoli footprints preserve earliest direct evidence of human-like bipedal biomechanics. *PLoS ONE* 5:e9769. doi:10.1371/journal.pone.0009769

[80] Semaw et al. (1997) 2.5-million-year-old stone tools from Gona, Ethiopia. *Nature* 385:333–336. doi:10.1038/385333a0

[81] Lepre et al. (2011) An earlier origin for the Acheulian. *Nature* 477:82–85. doi:10.1038/nature10372

[82] Plummer et al. (2023) Expanded geographic distribution and dietary strategies of the earliest Oldowan hominins and *Paranthropus*. *Science* 379:561–566. doi:10.1126/science.abo7452

[83] Harmand et al. (2015) 3.3-million-year-old stone tools from Lomekwi 3, West Turkana, Kenya. *Nature* 521:310–315. doi:10.1038/nature14464

[84] Archer, Aldeias and McPherron (2020) What is 'in situ'? A reply to Harmand et al. (2015). *J. Human Evolution* 142:102740. doi:10.1016/j.jhevol.2020.102740

[85] Rizal et al. (2020) Last appearance of *Homo erectus* at Ngandong, Java, 117,000–108,000 years ago. *Nature* 577:381–385. doi:10.1038/s41586-019-1863-2

CHECK THE EVIDENCE (4/5)

Sources you can inspect

[86] Brown et al. (1985) Early *Homo erectus* skeleton from west Lake Turkana, Kenya. *Nature* 316:788–792. doi:10.1038/316788a0

[87] Suwa et al. (2009) The *Ardipithecus ramidus* skull and its implications for hominid origins. *Science* 326:68. doi:10.1126/science.1175825

[88] Smithsonian Institution, *Brains* (human characteristics). humanorigins.si.edu/human-characteristics/brains

[89] VanSickle, Cofran and Hunt (2020) Did Neandertals have large brains? Factors affecting endocranial volume comparisons. *American Journal of Physical Anthropology* 173:768–775. doi:10.1002/ajpa.24124

[90] Dirks et al. (2017) The age of *Homo naledi* and associated sediments in the Rising Star Cave, South Africa. *eLife* 6:e24231. doi:10.7554/eLife.24231

[91] Berger and Hawks (2023) Revisiting the age of the Florisbad hominin material. *HOMO* 74:55–60. doi:10.1127/homo/2023/1541

[92] Ragsdale et al. (2023) A weakly structured stem for human origins in Africa. *Nature* 617:755–763. doi:10.1038/s41586-023-06055-y

[93] Harvati et al. (2019) Apidima Cave fossils provide earliest evidence of *Homo sapiens* in Eurasia. *Nature* 571:500–504. doi:10.1038/s41586-019-1376-z

[94] Gibbons (2017) Oldest members of our species discovered in Morocco. *Science* 356:993–994. doi:10.1126/science.356.6342.993

[95] Smithsonian Institution, *Homo sapiens*. humanorigins.si.edu/evidence/human-fossils/species/homo-sapiens

[96] Chimpanzee Sequencing and Analysis Consortium (2005) Initial sequence of the chimpanzee genome. *Nature* 437:69–87. doi:10.1038/nature04072

[97] Yoo et al. (2025) Complete sequencing of ape genomes. *Nature* 641:401–418. doi:10.1038/s41586-025-08816-3 (speciation times: Supplementary Note VI)

[98] Langergraber et al. (2012) Generation times in wild chimpanzees and gorillas suggest earlier divergence times in great ape and human evolution. *PNAS* 109:15716–15721. doi:10.1073/pnas.1211740109

[99] Moorjani et al. (2016) Variation in the molecular clock of primates. *PNAS* 113:10607–10612. doi:10.1073/pnas.1600374113

[100] Faria et al. (2014) The early spread and epidemic ignition of HIV-1 in human populations. *Science* 346:56–61. doi:10.1126/science.1256739

[101] Nishikimi et al. (1994) Cloning and chromosomal mapping of the human nonfunctional gene for L-gulono-γ-lactone oxidase, the enzyme for L-ascorbic acid biosynthesis missing in man. *J. Biological Chemistry* 269:13685–13688. doi:10.1016/S0021-9258(17)36884-9

[102] Ohta and Nishikimi (1999) Random nucleotide substitutions in primate nonfunctional gene for L-gulono-γ-lactone oxidase, the missing enzyme in L-ascorbic acid biosynthesis. *Biochimica et Biophysica Acta* 1472:408–411. doi:10.1016/S0304-4165(99)00123-3

[103] Mansueto et al. (2024) Conservation of a chromosome 8 inversion and exon mutations confirm common gulonolactone oxidase gene evolution among primates, including *H. neanderthalensis*. *J. Molecular Evolution* 92:266–277. doi:10.1007/s00239-024-10165-0

[104] International Human Genome Sequencing Consortium (2001) Initial sequencing and analysis of the human genome. *Nature* 409:860–921. doi:10.1038/35057062

[105] Johnson and Coffin (1999) Constructing primate phylogenies from ancient retrovirus sequences. *PNAS* 96:10254–10260. doi:10.1073/pnas.96.18.10254

[106] US National Human Genome Research Institute (2005) *Scientists analyze chromosomes 2 and 4*. genome.gov/13514624/2005-release-scientists-analyze-chromosomes-2-and-4

[107] Hillier et al. (2005) Generation and annotation of the DNA sequences of human chromosomes 2 and 4. *Nature* 434:724–731. doi:10.1038/nature03466

[108] Avarello et al. (1992) Evidence for an ancestral alphoid domain on the long arm of human chromosome 2. *Human Genetics* 89:247–249. doi:10.1007/BF00217134

[109] NHM, *Who were the Neanderthals?* nhm.ac.uk/discover/who-were-the-neanderthals.html

[110] Sümer et al. (2025) Earliest modern human genomes constrain timing of Neanderthal admixture. *Nature* 638:711–717. doi:10.1038/s41586-024-08420-x

[111] NHM news (2024) *Neanderthals and Homo sapiens interbred within the past 50,000 years*. nhm.ac.uk/discover/news/2024/december/neanderthals-homo-sapiens-interbred-within-past-50000-years.html

[112] Smithsonian Institution, *Ancient DNA and Neanderthals*. humanorigins.si.edu/evidence/genetics/ancient-dna-and-neanderthals

CHECK THE EVIDENCE (5/5)

Sources you can inspect

[113] Larena et al. (2021) Philippine Ayta possess the highest level of Denisovan ancestry in the world. *Current Biology* 31:4219–4230. doi:10.1016/j.cub.2021.07.022

[114] NHM, *Cheddar Man: Mesolithic Britain's blue-eyed boy*. nhm.ac.uk/discover/cheddar-man-mesolithic-britain-blue-eyed-boy.html

[115] Clarkson et al. (2017) Human occupation of northern Australia by 65,000 years ago. *Nature* 547:306–310. doi:10.1038/nature22968

[116] O'Connell et al. (2018) When did *Homo sapiens* first reach Southeast Asia and Sahul? *PNAS* 115:8482–8490. doi:10.1073/pnas.1808385115

[117] Chen et al. (2019) A late Middle Pleistocene Denisovan mandible from the Tibetan Plateau. *Nature* 569:409–412. doi:10.1038/s41586-019-1139-x

[118] Skoglund and Reich (2016) A genomic view of the peopling of the Americas. *Current Opinion in Genetics & Development* 41:27–35. doi:10.1016/j.gde.2016.06.016

[119] Li and Durbin (2011) Inference of human population history from individual whole-genome sequences. *Nature* 475:493–496. doi:10.1038/nature10231

[120] Poznik et al. (2013) Sequencing Y chromosomes resolves discrepancy in time to common ancestor of males versus females. *Science* 341:562–565. doi:10.1126/science.1237619

[121] Swamidass (2019) *The Genealogical Adam and Eve: The Surprising Science of Universal Ancestry*. IVP Academic.

[122] Swamidass, S. J. (2019) The misunderstood science of genetic bottlenecks. *PeacefulScience*. peacefulscience.org/prints/misunderstood-bottleneck/ (the author's own summary: lineage-time inference rules out a sole founding pair back to about 500,000 years).

[123] Pew Research Center (2015) *Elaborating on the views of AAAS scientists, issue by issue*. Survey of 3,748 AAAS scientists and 2,002 US adults, 2014. pewresearch.org/internet/2015/07/23/elaborating-on-the-views-of-aaas-scientists-issue-by-issue/

[124] Antimicrobial Resistance Collaborators [Murray et al.] (2022) Global burden of bacterial antimicrobial resistance in 2019: a systematic analysis. *Lancet* 399:629–655. doi:10.1016/S0140-6736(21)02724-0

[125] Xenophanes on fossil shells: Hippolytus, *Refutation of All Heresies* I.14 (DK 21 A33). Aristotle, *Metaphysics* I.1, 980a21, tr. W. D. Ross (1924). Lucretius, *De rerum natura*, Book V.

[126] Augustine of Hippo (c. 401–415) *De Genesi ad litteram* (*The Literal Meaning of Genesis*), book IV.

[127] Ussher, J. (1650) *Annales Veteris Testamenti.* The date 4004 BCE was printed in the margins of many English Bibles from 1701.

[128] Jouzel et al. (2007) Orbital and millennial Antarctic climate variability over the past 800,000 years. *Science* 317:793–796. doi:10.1126/science.1141038

[129] Sommeria (2017) Foucault and the rotation of the Earth. *Comptes Rendus Physique* 18:520–525. doi:10.1016/j.crhy.2017.11.003

[130] European Space Agency, *A history of astrometry, Part II: Telescope ignites the race to measure stellar distances* (Bessel, 1838, 61 Cygni). sci.esa.int/web/gaia/-/53197-seeing-and-measuring-farther

[131] Southerton, S. G. (2004) *Losing a Lost Tribe: Native Americans, DNA, and the Mormon Church*. Signature Books.

[132] Deseret News (2007) Debate renewed with change in Book of Mormon introduction, 8 November. deseret.com/2007/11/8/20052445/debate-renewed-with-change-in-book-of-mormon-introduction/

[133] Talarico and Rubin (2003) Confidence, not consistency, characterizes flashbulb memories. *Psychological Science* 14:455–461. doi:10.1111/1467-9280.02453

[134] Feynman, R. P. (1974) Cargo cult science. Caltech commencement address; *Engineering and Science* 37(7):10–13.

[135] NHM, *Piltdown Man*. nhm.ac.uk/our-science/services/library/collections/piltdown-man.html

[136] NASA, *Supernova 1987A*. science.nasa.gov/asset/hubble/supernova-1987a

[137] Humanists International (2022) *The Amsterdam Declaration 2022*. humanists.international/what-is-humanism/the-amsterdam-declaration/

[138] Brumm et al. (2021) Oldest cave art found in Sulawesi. *Science Advances* 7:eabd4648. doi:10.1126/sciadv.abd4648

[139] Oktaviana et al. (2024) Narrative cave art in Indonesia by 51,200 years ago. *Nature* 631:814–818. doi:10.1038/s41586-024-07541-7

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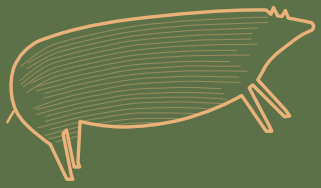
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About the cover. The pig is traced from a painting in Leang Tedongnge cave on Sulawesi, Indonesia, found in December 2017 by the archaeologist Basran Burhan. It is 136 cm long, painted in red ochre beside two hand stencils. Mineral crust that later formed over the paint has been dated by uranium-series to at least 45,500 years ago, so the painting is at least that old: one of the oldest known pictures of an animal. [138] Nearby, at Leang Karampuang, a painted scene of human-like figures and a pig has since been dated to at least 51,200 years ago. [139]

Facts cross-checked against the Natural History Museum, London, and the Smithsonian Institution. Page 60 lists the museum page behind each claim. Neither institution produced or endorsed this booklet. Some images were generated with AI tools; see page 59 for credits.

George Philippou
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